

EEE 443
ARTIFICIAL NEURAL NETWORKS

ÖMER MORGÜL

BİLKENT UNIVERSITY
DEPARTMENT OF ELECTRICAL ENG.
ANKARA
TURKEY

morgul@ee.bilkent.edu.tr

INTRODUCTION

- Basics of Neural Networks
- Our Aim
- Applications
- Neuron Structure
- Signal Generation/Propagation
- Simple Neuron Models/Applications
- Brief History

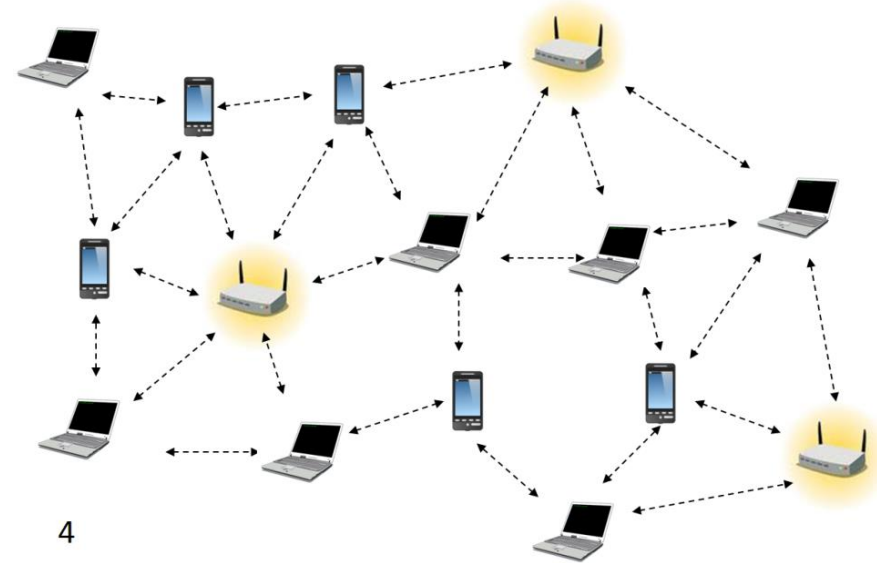
Neural: $\implies$. Neuron. Basic unit (cell) of nervous systems.

Network: $\implies$. An arbitrary interconnection of predetermined blocks

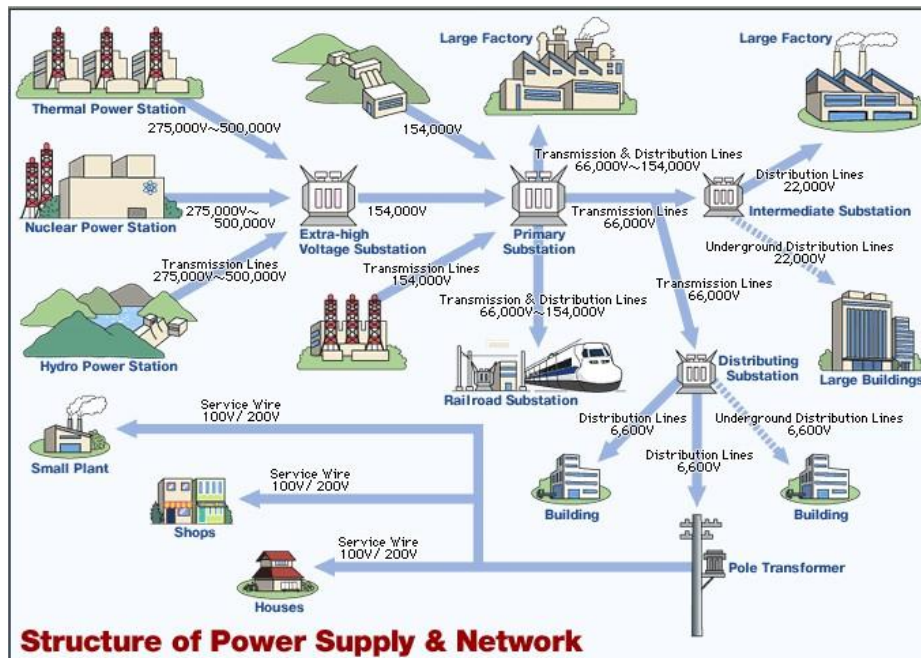
- Arbitrary interconnection ?
 - Computer Networks
 - Communication Networks
 - Social Networks
 - Power Networks
- All networks are configured to perform a specific task (decision making, object detection, classification etc.)



computer/communication network

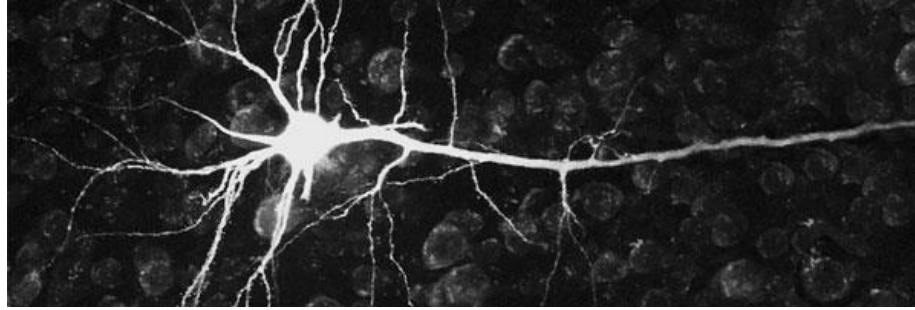


4

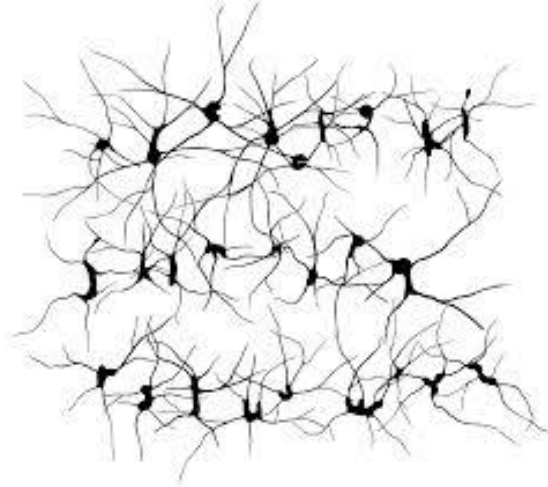


power network

- Basic building block (unit) of a Neural Network is a neuron.



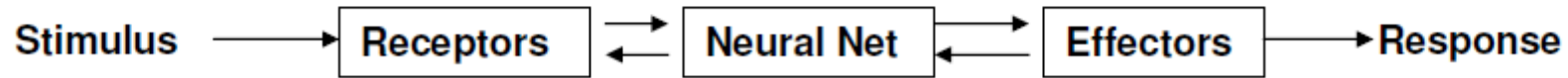
- **Neural Network:** Arbitrary connections of such neuron units.



mathematical model ?

The Nervous System

The human nervous system can be broken down into three stages that can be represented in block diagram form as



(adapted from Arbib, 1987)

The **receptors** convert stimuli from the external environment into electrical impulses that convey information to the neural net (brain)

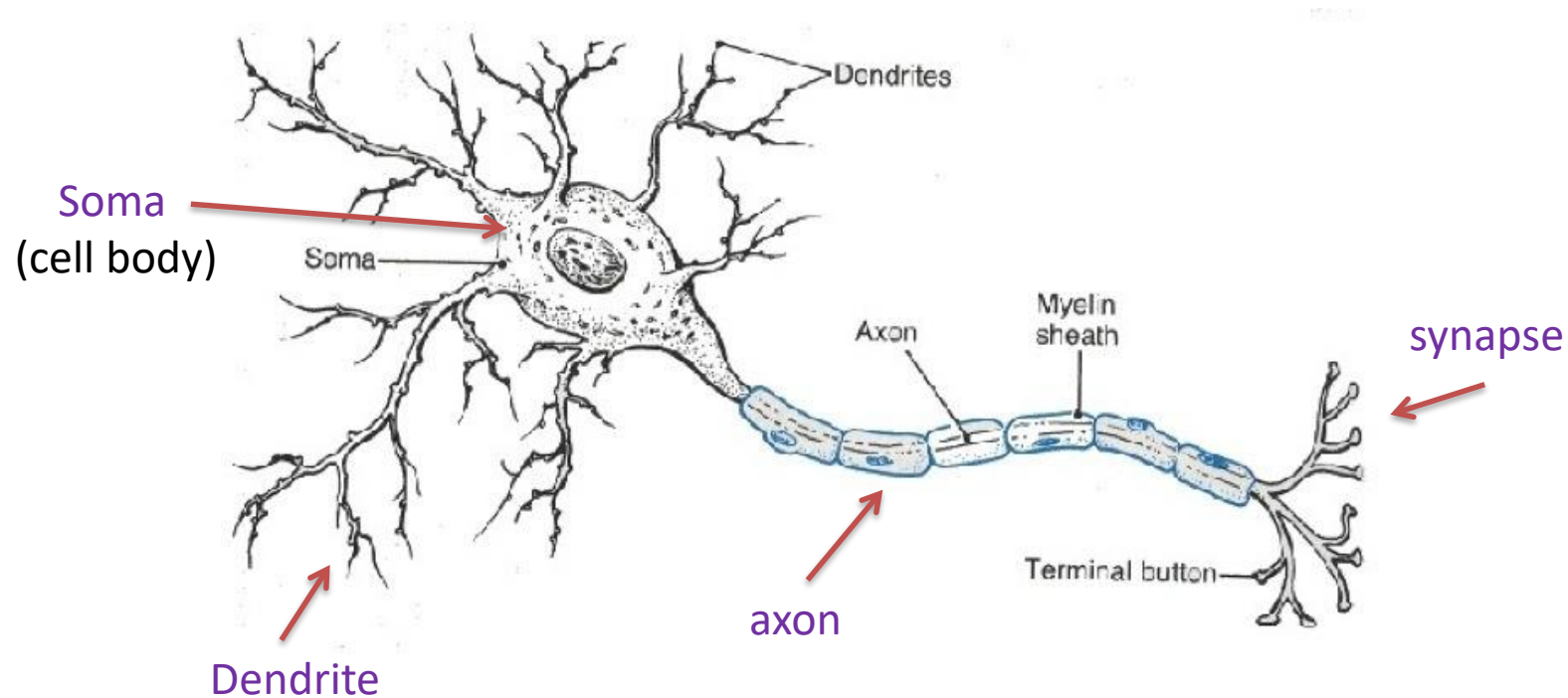
The **effectors** convert electrical impulses generated by the neural net into responses as system outputs

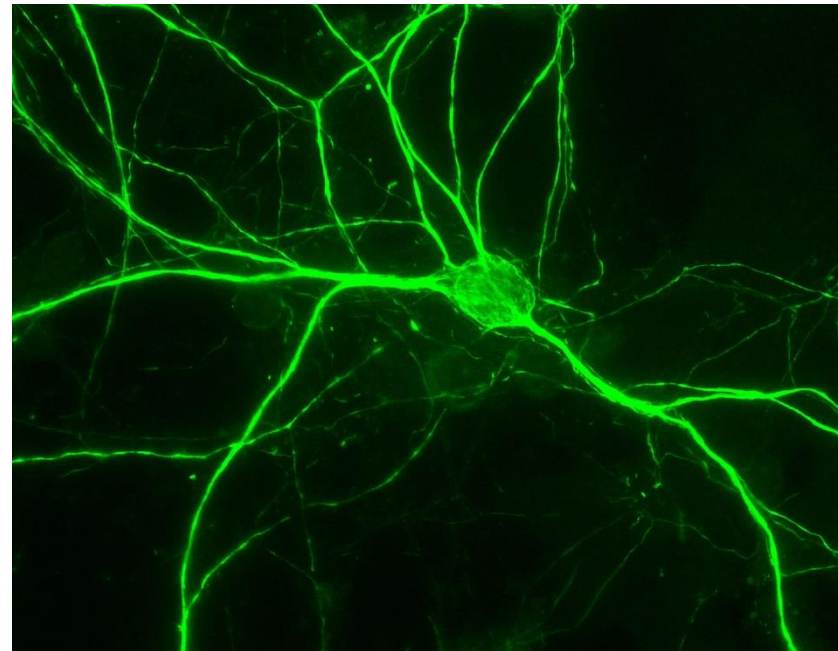
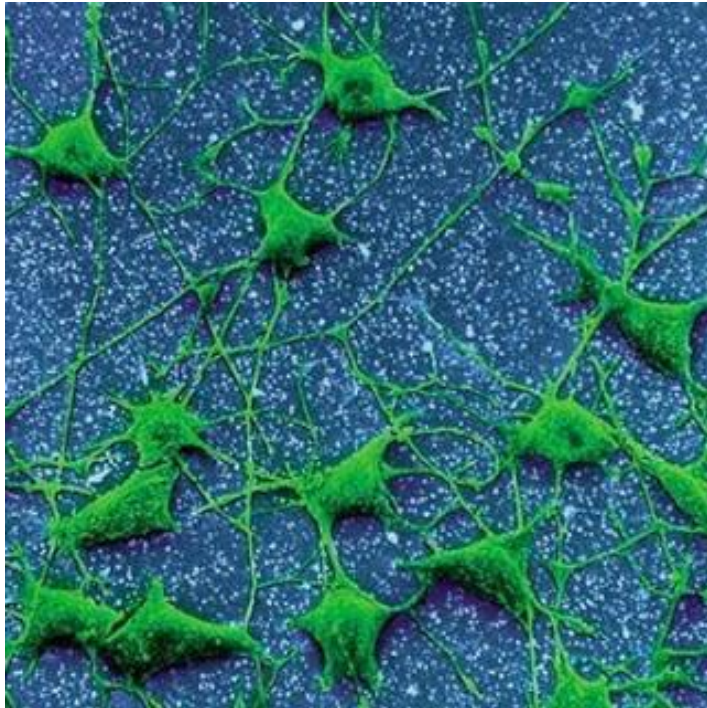
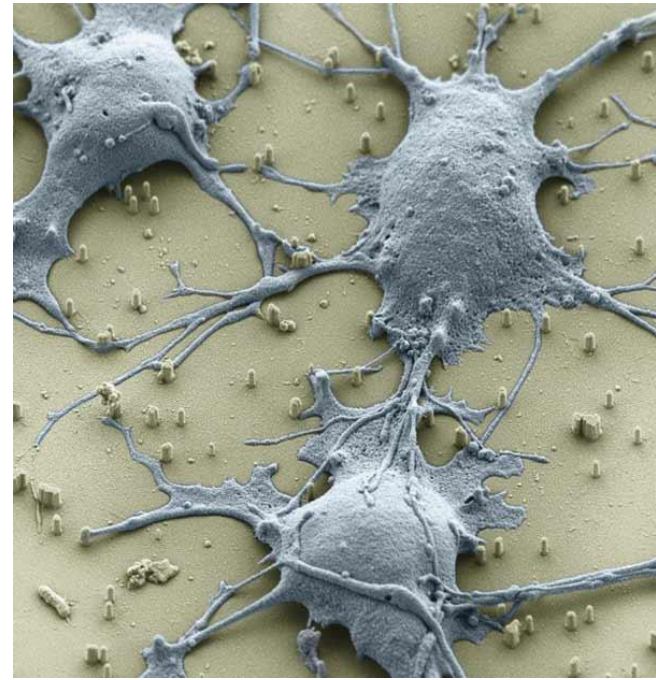
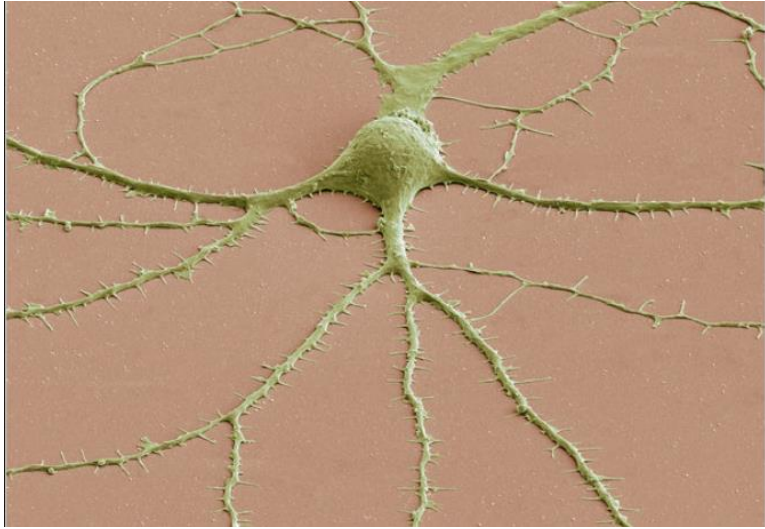
The **neural net (brain)** continually receives information, perceives it and makes appropriate decisions.

The flow of information is represented by arrows – feedforward and feedback

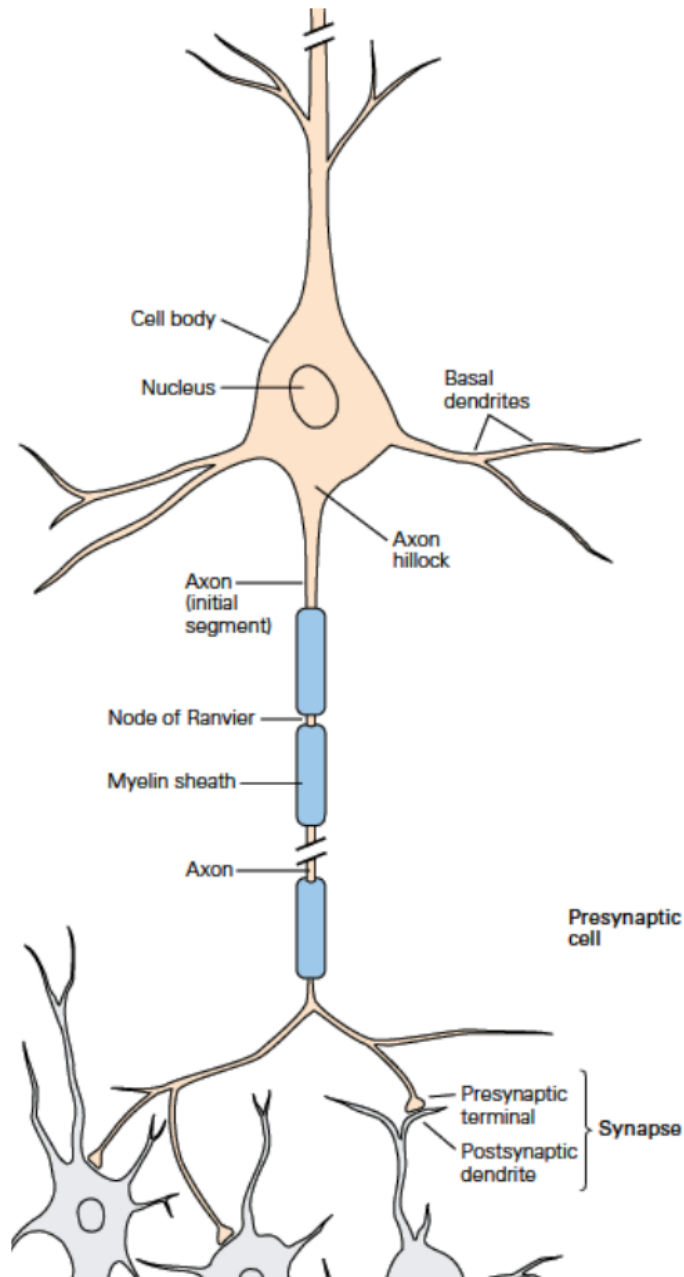
● Types of Neurons :

- Interneuron. From Neurons to Neurons.
 - Sensory Neurons. connected to the sensory organs
 - Motor Neurons. Connected to muscles.
-
- different structures, sizes, shapes, same components :





Structure of a Neuron



- Dendrites:
 - Collects potentials from other neurons
 - Radius: a few microns
- Soma:
 - Collects dendrite potentials
 - 10-80 microns
- Axons:
 - Transmission lines
 - Radius: micron to mm
 - Length: micron to m
- Synapse:
 - 200 nm gap
 - Neurotransmitters

- Dendrites. Input ports. Collects “data” (potential) from nearby neurons. Cylindrical shape. Radius $\sim$ a few micron.
- Soma. Main cell body. Collects the potential received from dendrites.
Size : $\sim 10 - 80$ microns.
- Axon. Transmission line. Cylindrical shape. Radius : $\sim 0.2\mu - 1mm$.
Length : $\sim$ a few μ - 1 m!. covered by myelin sheet
- Synapses. Transmits the data to the next neuron
Radius $\sim$ a few micron.
- Total Length : 10μ (for interneurons) - 1 m (for motor neurons).
- Synaptic Gap. Between a synapse and dendrite. ~ 200 nm.

Information flow through neurons

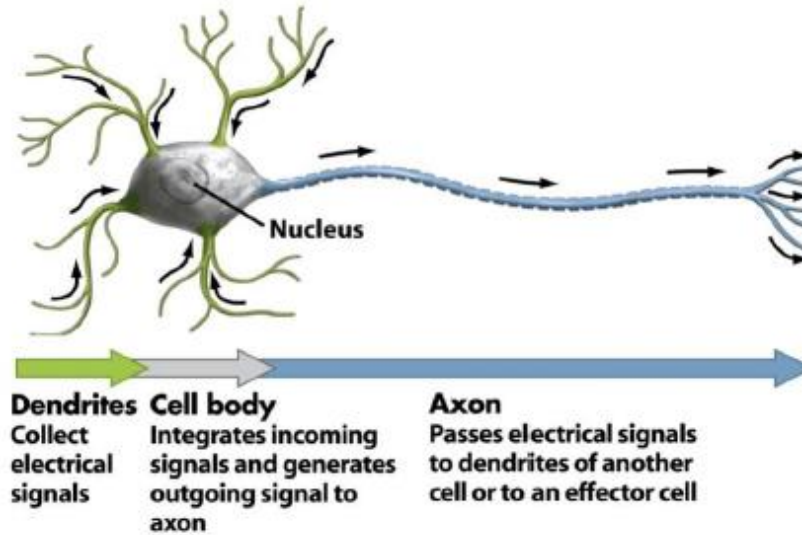
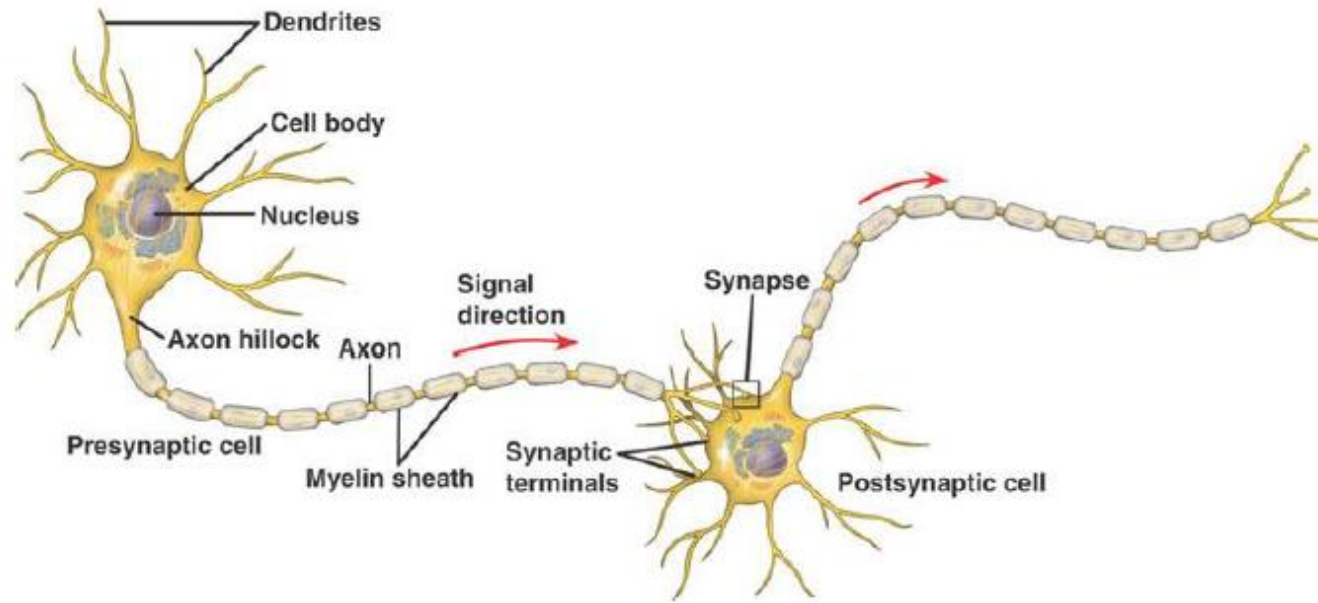
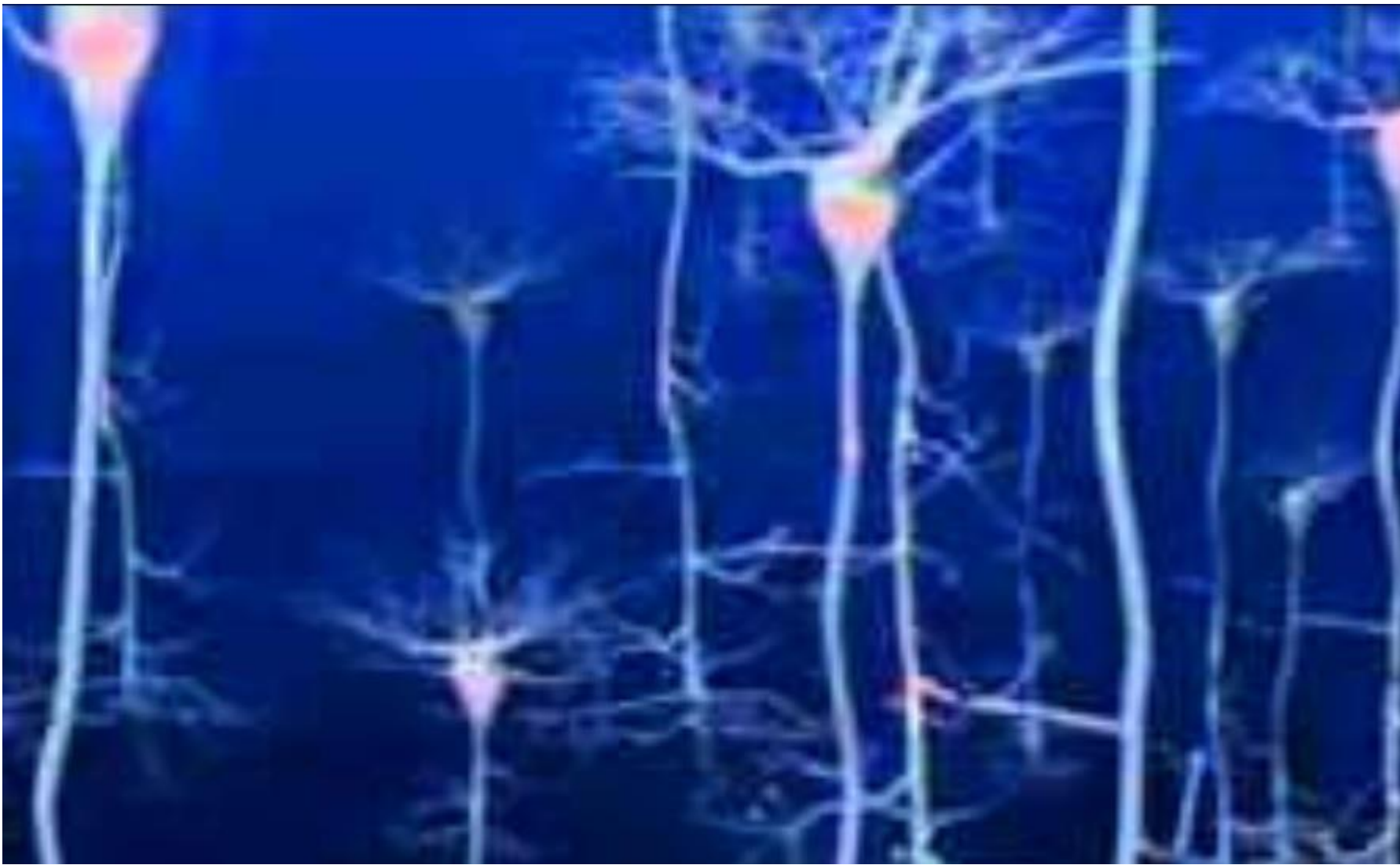


Figure 45-2b Biological Science, 2/e
© 2005 Pearson Prentice Hall, Inc.

- **Information Flow** : One direction only. (Neuron 1 : Dendrites → Soma → Axon → Synapses → (Neuron 2 : Dendrites).



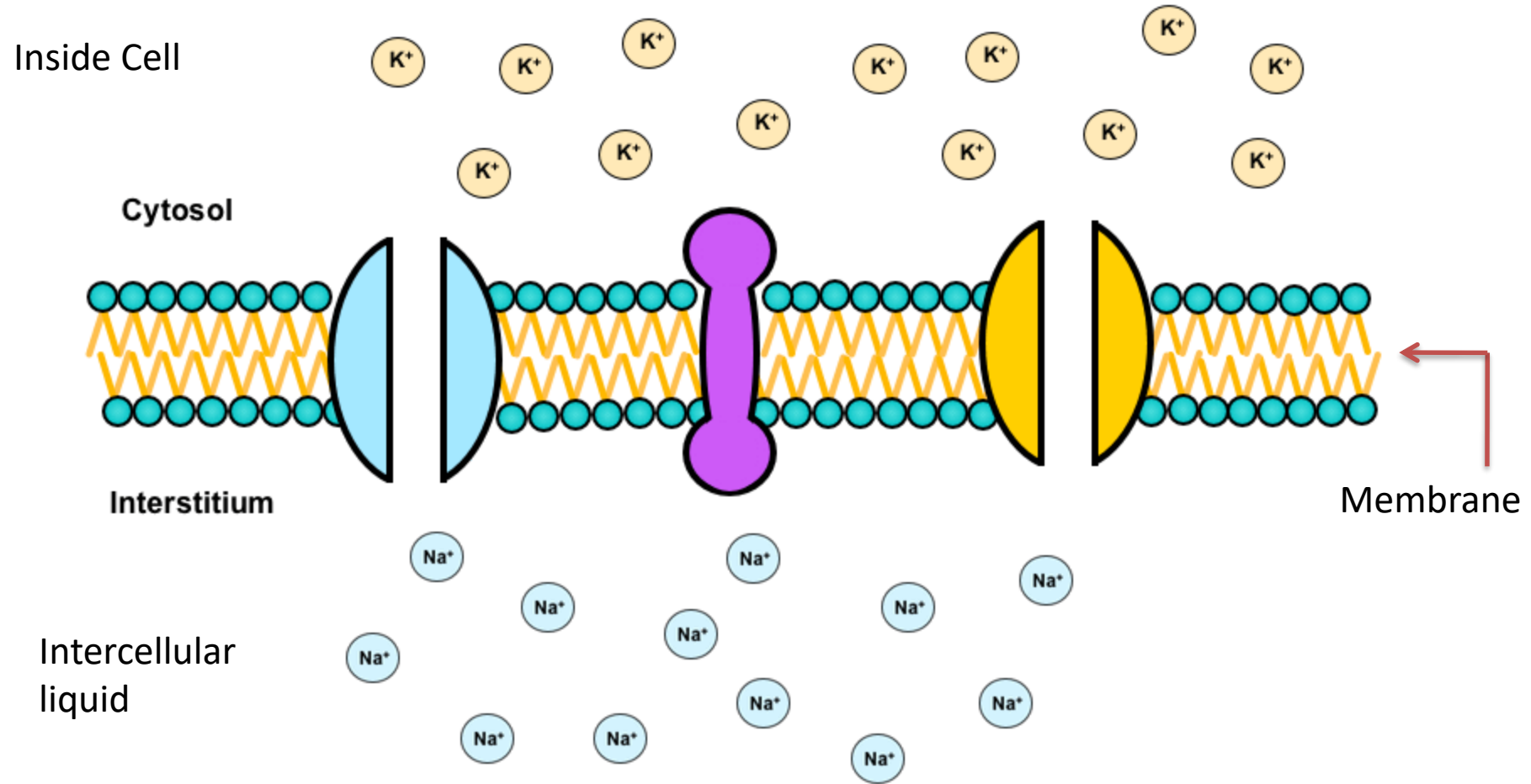


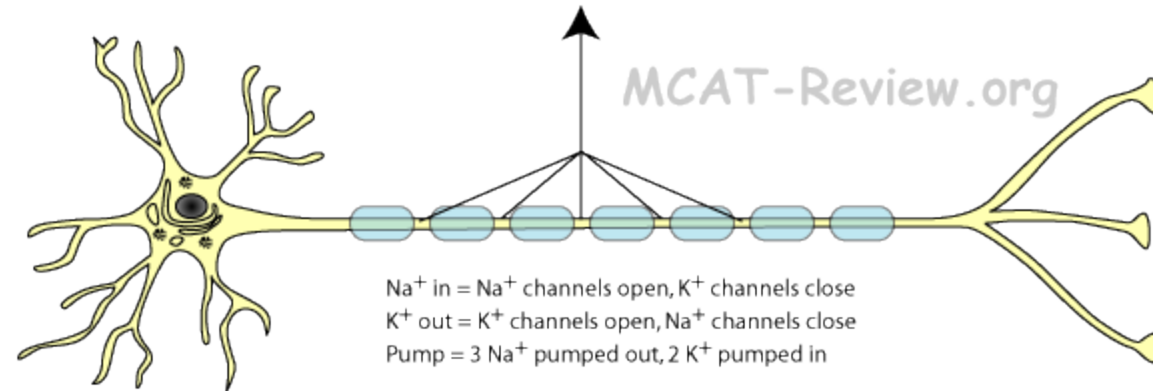
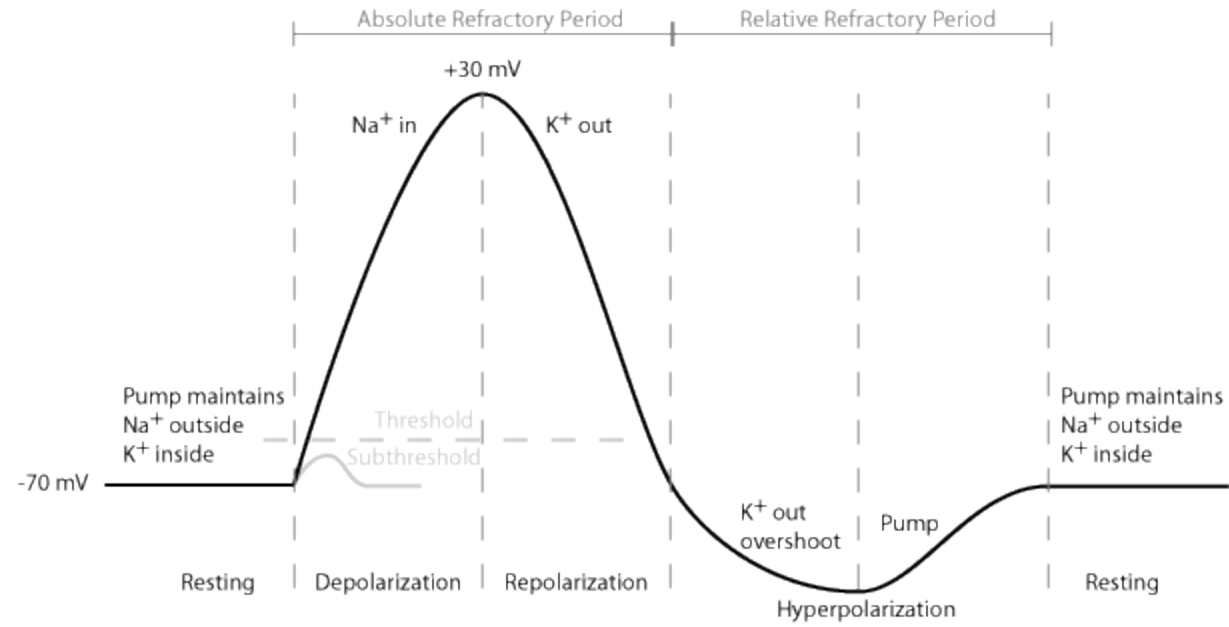
http://www.youtube.com/watch?v=GIQqp6_PG6k

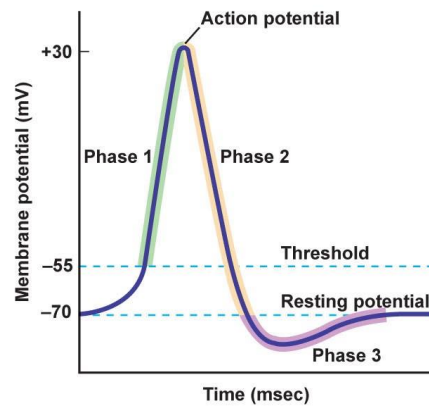
- Neutrally, cell membrane has more resistance (less conductance/permeability) to Na ions as compared to K ions (ratio ~ 1000). As a result, there are more K ions and less Na ions inside as compared to outside. Similarly, there are more Cl ions outside as compared to inside. Organic ions are too large to pass the cell membrane too often.

- **Result :** A chemical balance occurs. Around $70 - 100 \text{ mV}$ potential occurs between inside and outside of the cell, inside being more negative. -70 mV is more common. This is called **Resting Potential**.

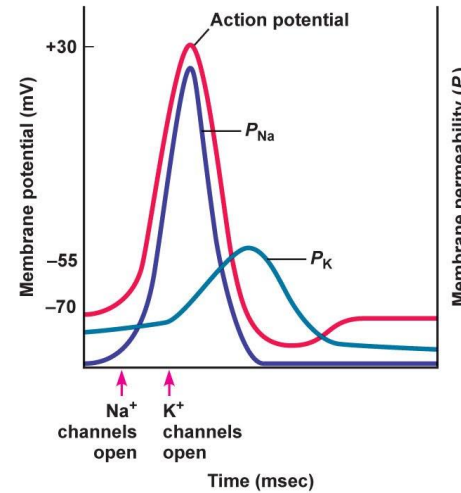
Depolarizace





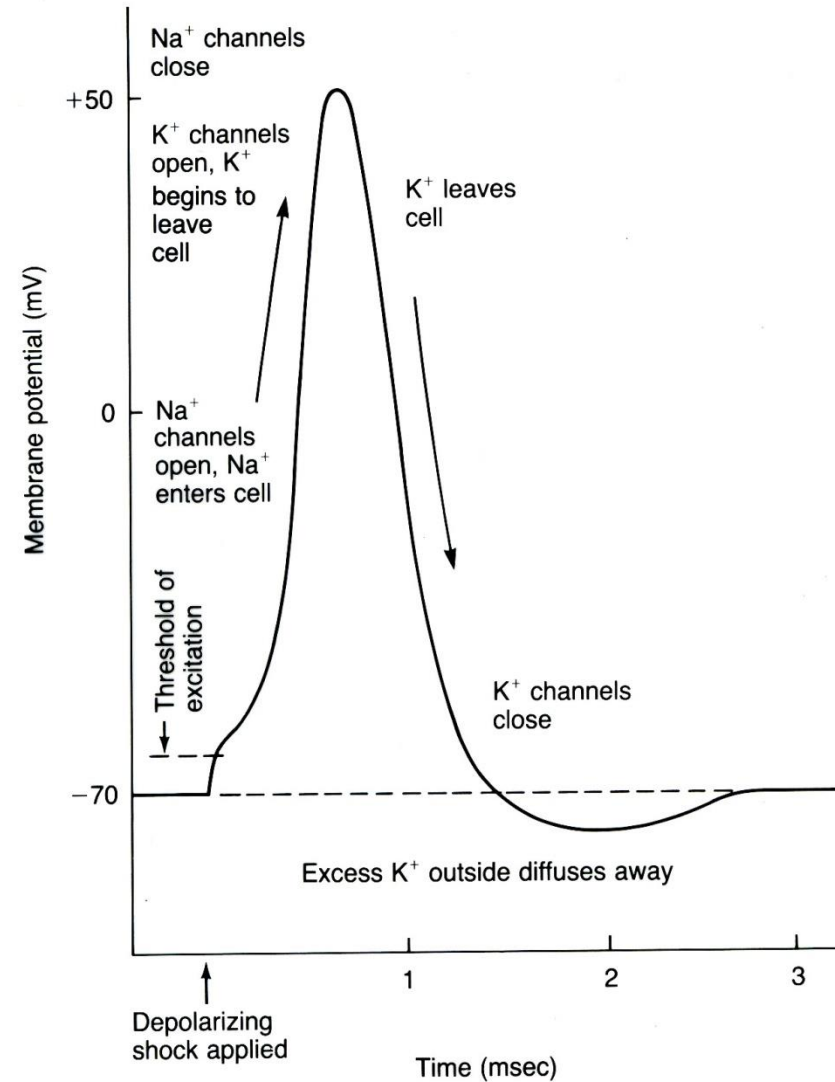


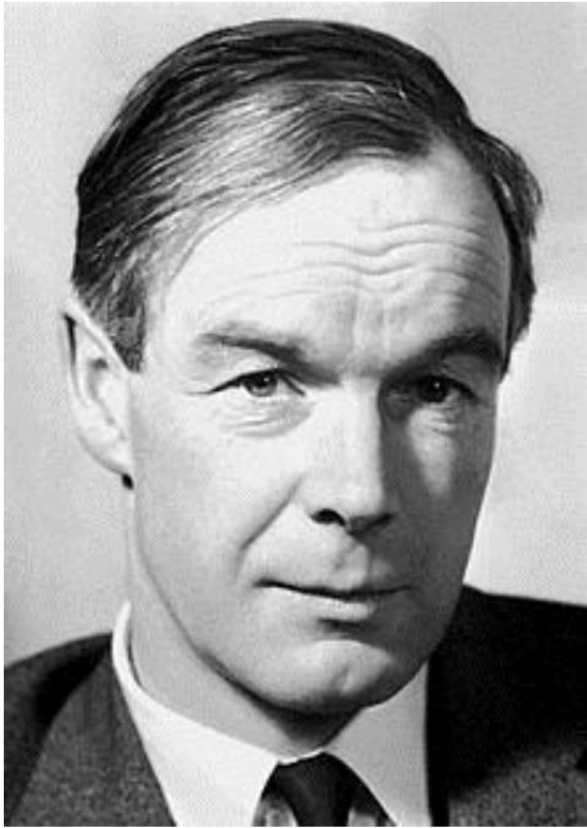
(a) Three phases of an action potential

(b) Permeability changes for Na^+ and K^+ during an action potential

© 2011 Pearson Education, Inc.

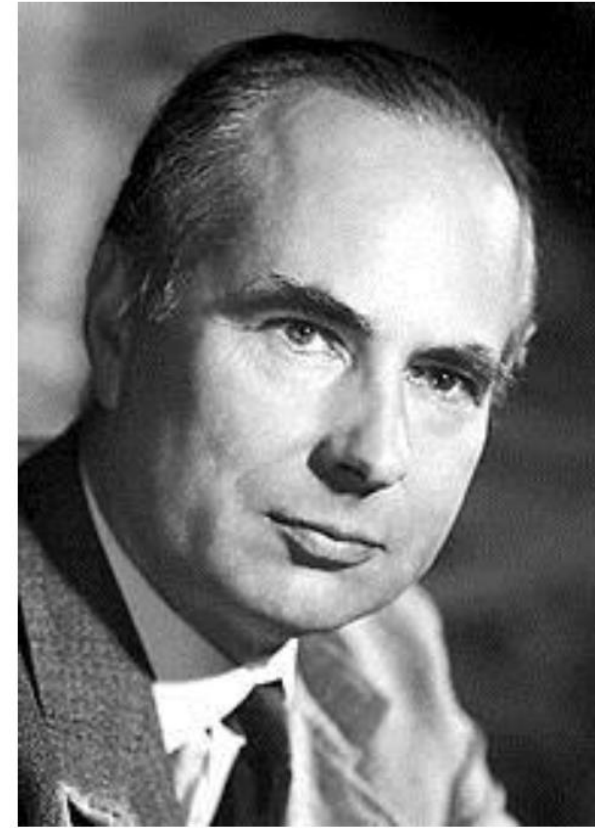
- input through dendrites
- soma potential changes
- when a threshold is passed, resistivity wrt Na/K drops
- No input from outside
- Then everything is back to resting...
- Action potential...





Alan Lloyd Hodgkin

1914-1998

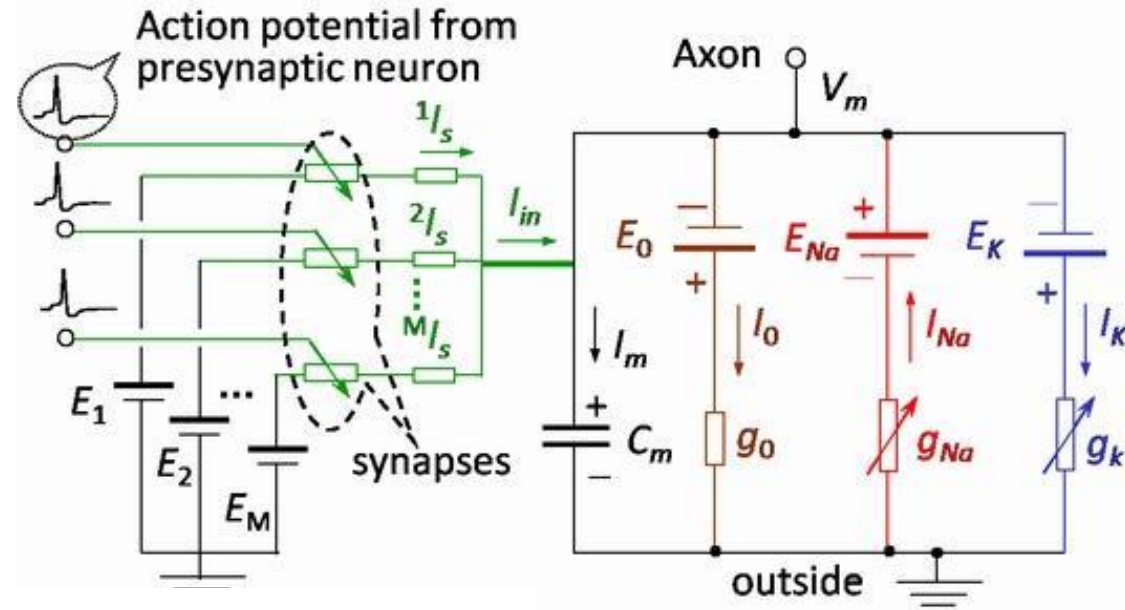


Andrew Huxley

1917-2012

1963 Nobel Price in Medicine

Hodgkin-Huxley Model



Hodgkin-Huxley equations

$$C_m \frac{dV}{dt} = -g_L(V - V_L) - \bar{g}_{Na} m^3 h (V - V_{Na}) - \bar{g}_K n^4 (V - V_K)$$

$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m$$

$$\frac{dh}{dt} = \alpha_h(V)(1 - h) - \beta_h(V)h$$

$$\frac{dn}{dt} = \alpha_n(V)(1 - n) - \beta_n(V)n$$

m : potassium channel activation

h : sodium channel activation

n : sodium channel inactivation

A QUANTITATIVE DESCRIPTION OF MEMBRANE
CURRENT AND ITS APPLICATION TO CONDUCTION
AND EXCITATION IN NERVE

BY A. L. HODGKIN AND A. F. HUXLEY

From the Physiological Laboratory, University of Cambridge

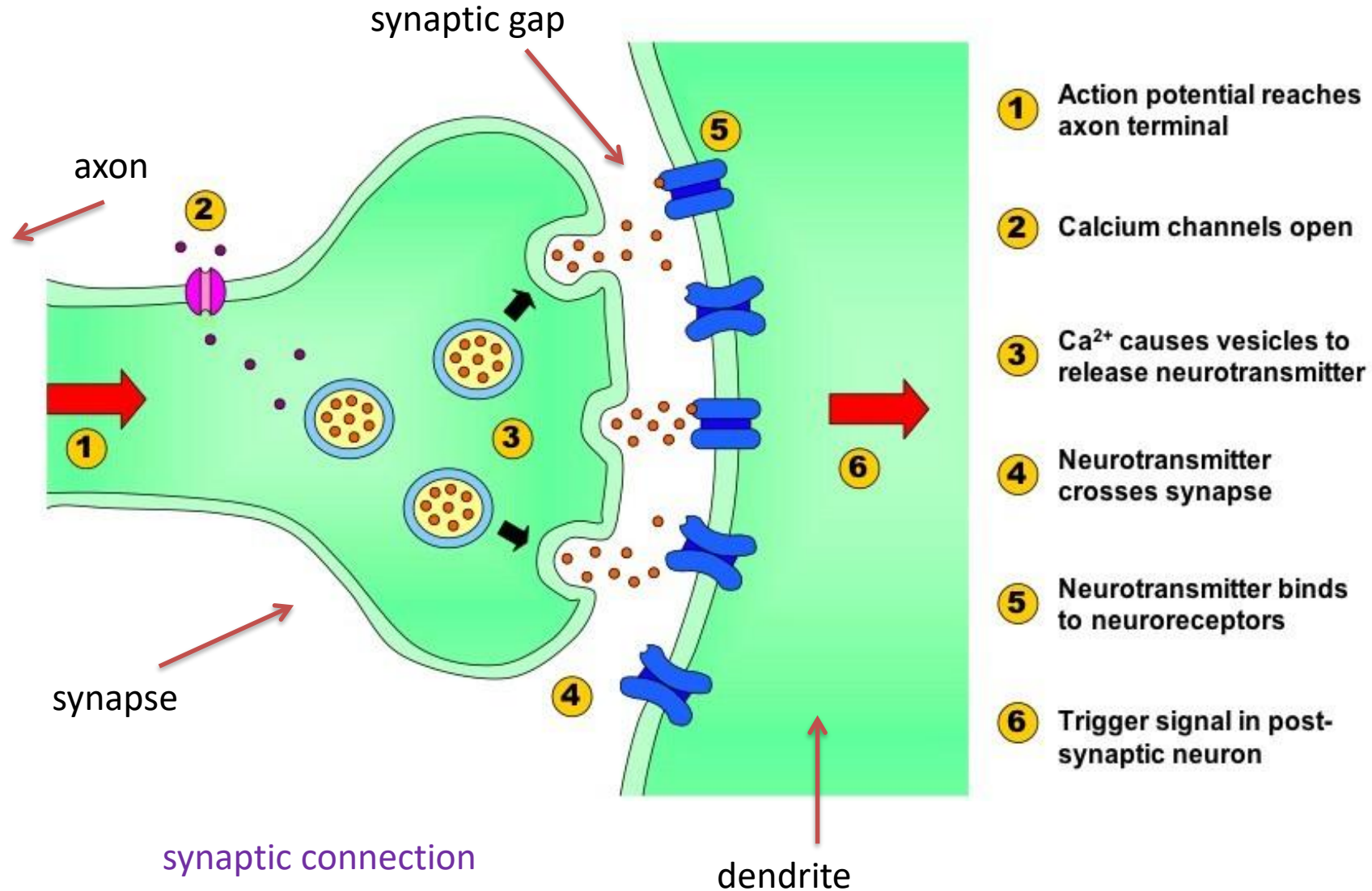
(Received 10 March 1952)

This article concludes a series of papers concerned with the flow of electric current through the surface membrane of a giant nerve fibre (Hodgkin, Huxley & Katz, 1952; Hodgkin & Huxley, 1952 *a-c*). Its general object is to discuss the results of the preceding papers (Part I), to put them into mathematical form (Part II) and to show that they will account for conduction and excitation in quantitative terms (Part III).

PART I. DISCUSSION OF EXPERIMENTAL RESULTS

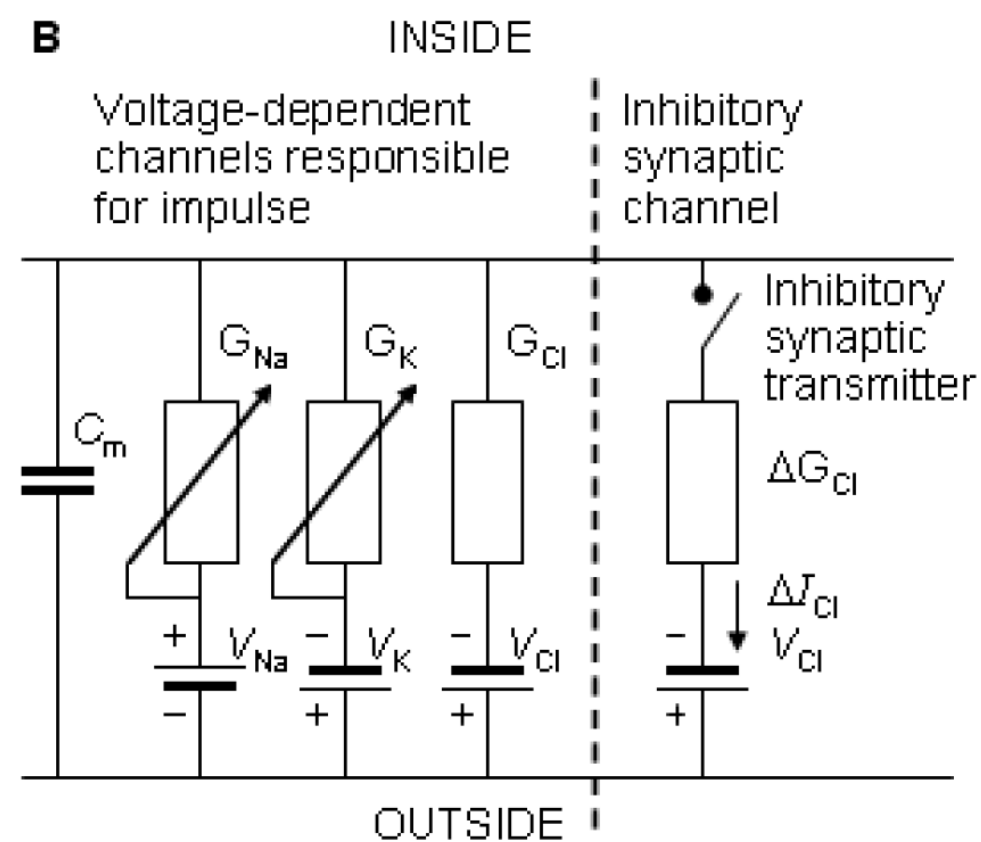
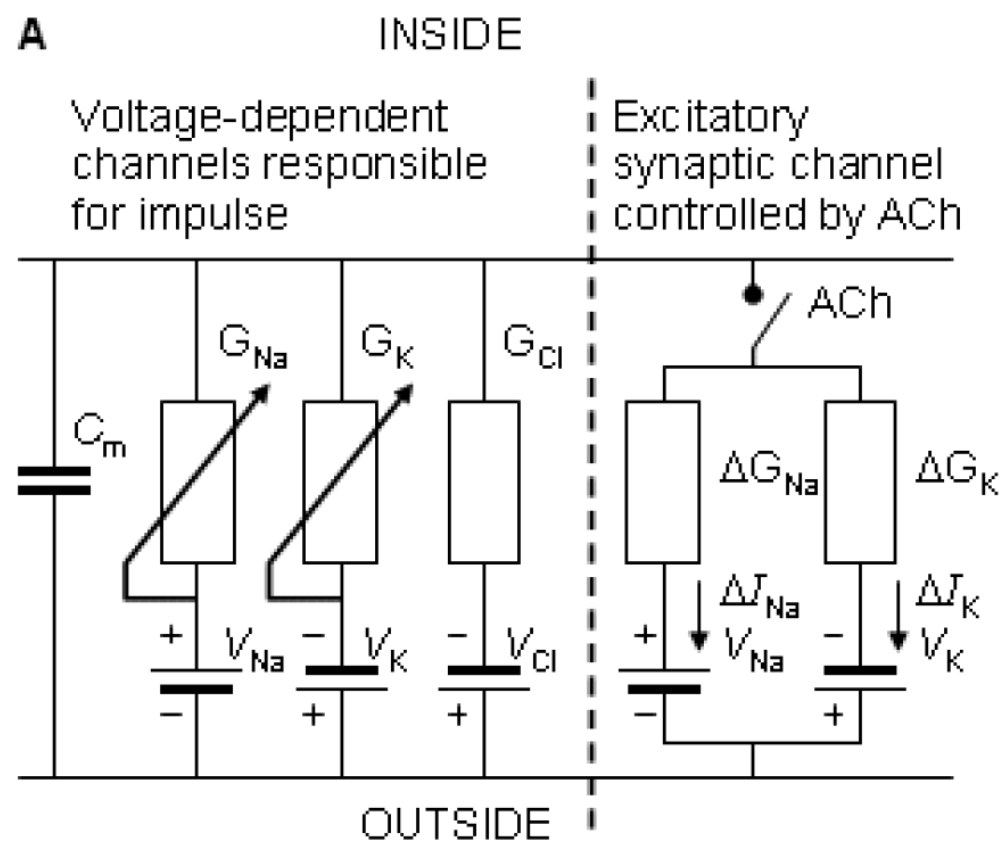
The results described in the preceding papers suggest that the electrical behaviour of the membrane may be represented by the network shown in Fig. 1. Current can be carried through the membrane either by charging the membrane capacity or by movement of ions through the resistances in parallel with the capacity. The ionic current is divided into components carried by sodium and potassium ions (I_{Na} and I_K), and a small 'leakage current' (I_l) made up by chloride and other ions. Each component of the ionic current is determined by a driving force which may conveniently be measured as an electrical potential difference and a permeability coefficient which has the dimensions of a conductance. Thus the sodium current (I_{Na}) is equal to the sodium conductance (g_{Na}) multiplied by the difference between the membrane potential (E) and the equilibrium potential for the sodium ion (E_{Na}). Similar equations apply to I_K and I_l and are collected on p. 505.

● Signal Transmission :

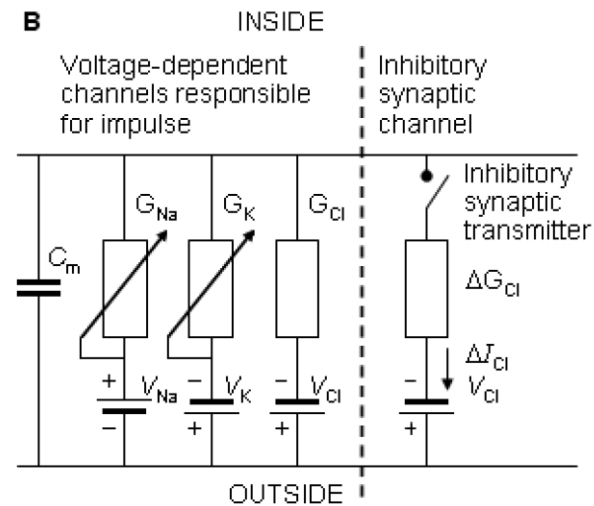
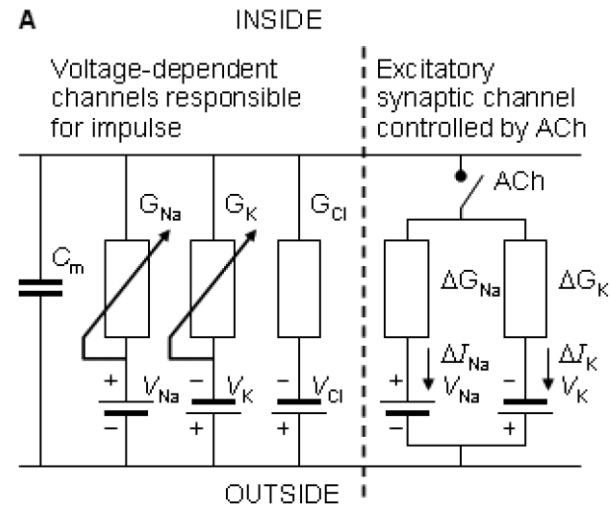


- When action potential reaches the synapses, they release chemical substances what is called **neurotransmitters** (e.g. 10^{-17} mol. acetylcholin per impulse). These neurotransmitters reaches the next dendrite around 0.5 *msec*. Their role is to change the permeability of the other side with respect to Na, K ions (to increase the potential → **excitatory synapse**), or to Cl ions (to decrease the potential → **inhibitory synapse**). This behaviour is fixed in the early stages of life after some learning process. But the synapses may have different **strengths** which affect the amount of released neurotransmitters.

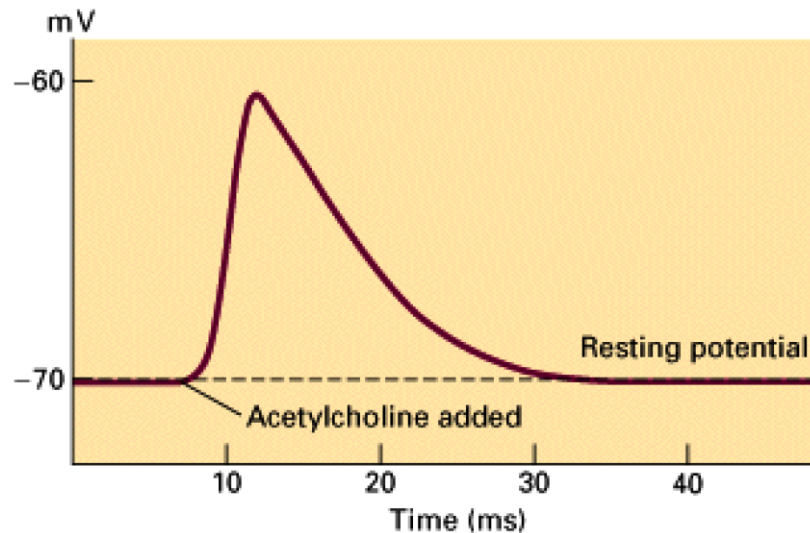
Excitatory versus Inhibitory Synapses



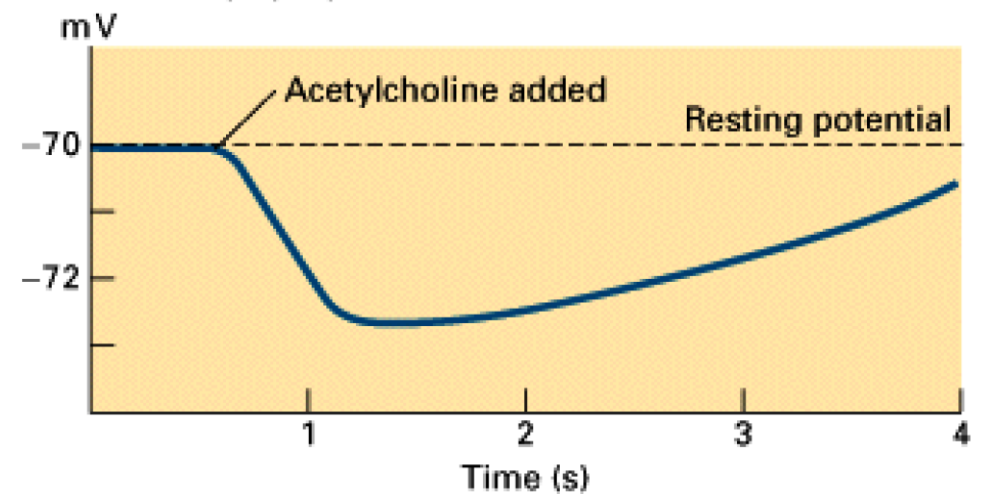
Excitatory versus Inhibitory Synapses



(a) Excitatory synapse



(b) Inhibitory synapse





Henry Hallett Dale

1875-1968



Otto Loewi

1873-1961

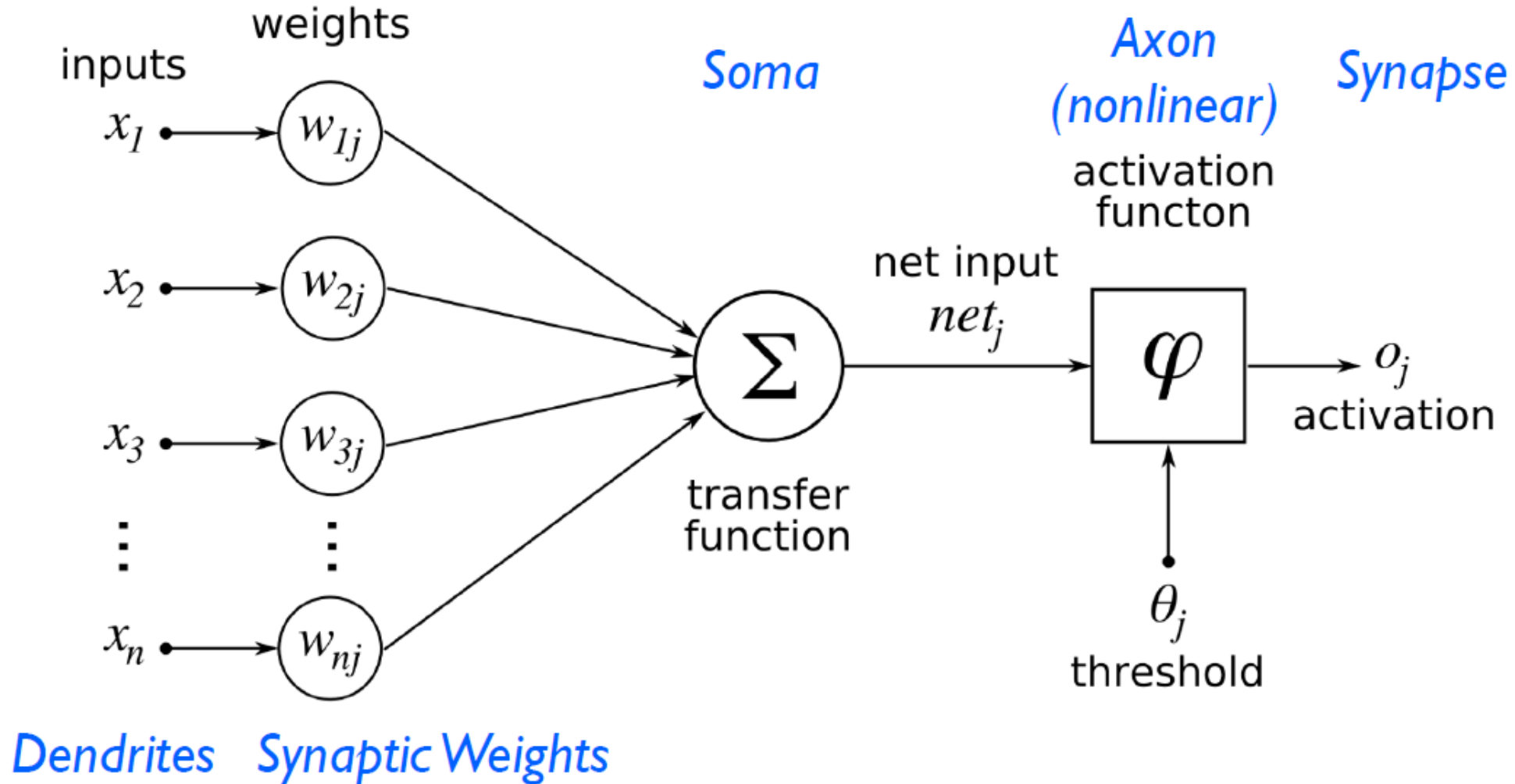
1936 Nobel Price in Medicine

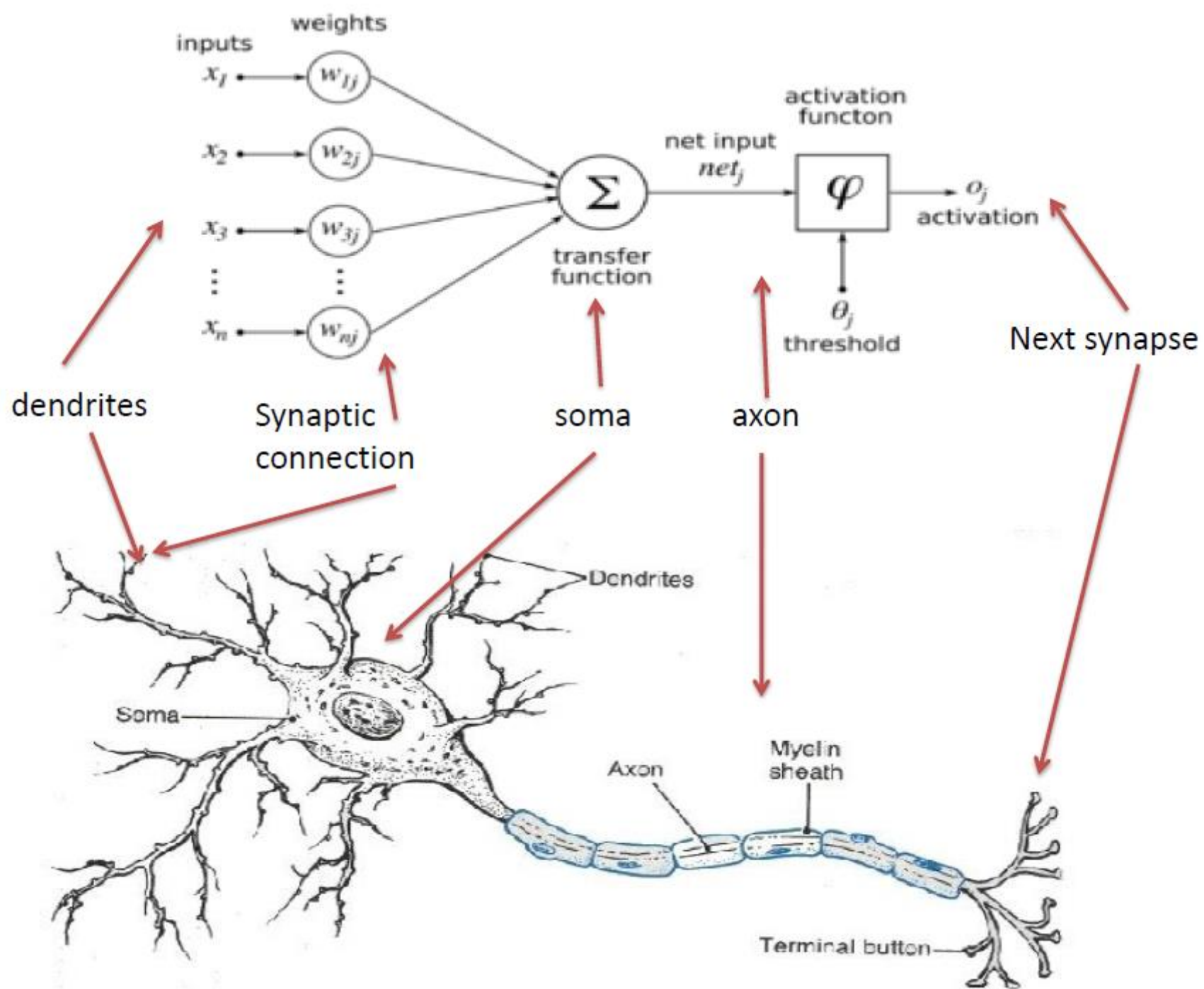
Definition of an artificial neural network

An information processing system that has been developed as a generalization of mathematical models of human cognition or neurobiology, based on the assumptions that

- Information processing occurs at many **simple elements called neurons**.
- Signals are passed between neurons over **connection links**.
- Each connection link has an **associated weight**, which typically multiplies the signal transmitted.
- Each neuron applies an **activation function** (usually non-linear) to its net input (sum of weighted input signals) to determine its output signal.

McCulloch-Pitts Neuron





- **The speed of transmission** : $0.5 - 2 \text{ m/sec}$ in interneurons.
- Any two brain cell can communicate in about $20 - 40 \text{ msec}$.
- In long axons, the speed of transmission can go as high as 100 m/sec .
- The frequency of spikes : $\leq 1 \text{ KHz}$. (signal duration $\sim 1 \text{ msec}$).

- **Is the information processed digitally ?**

Not exactly. may also be coded in the frequency

- **How many neurons are there?**

$\sim 3 \times 10^{10}$ in the brain, $\sim 10^{11}$ in the nervous system.

$\sim 10^4$ synaptic connections. $\rightarrow \sim 10^{15}$ total

$\sim 10^{11}$ processors with $\sim 10^{15}$ connections!

- Energetic efficiency of computers : 10^{-6} Joules per operation per second.
- Energetic efficiency of neurons : 10^{-16} Joules per operation per second!
- Speed of computers : $\sim 10^{-9}$ second per operation.
- Speed of neurons : $\sim 10^{-3}$ second per operation.
- Classical computers are useful in problems whose solutions have well defined algorithms which can be implemented serially. $\rightarrow$ multiplication, linear programming, etc...
- Neural Networks are more efficient in cognitive related tasks which does not have well defined algorithms. $\rightarrow$ pattern recognition, problem solving, motion coordination and planning etc..

Brains and Computers

Both systems....

process information

represent signals as differences in electrical potential

convey signals on “wires”

use active devices formed by thin energy barriers --> gain

Differences:

	<i>Brains</i>	<i>Computers</i>
matter	squishy cells	inorganic matter
power source	metabolic biochemistry	transformers and power mains
signal levels	100 mV	5 V
temporal resolution	milliseconds	nanoseconds
robustness to damage	good	poor
power consumption	10^{-12} watt/cell	10^{-5} watt/cell

Brains vs. Computers

Processing elements: There are 10^{11} neurons in the brain, compared with 5×10^9 transistors in an 18-core CPU

Processing speed: 100 Hz for the brain compared to 10^9 Hz for the computer

Style of computation: The brain computes in parallel and distributed mode, whereas the computer mostly serially and centralized.

Fault tolerant: The brain is fault tolerant, whereas the computer is not

Adaptive: The brain learns fast, whereas the computer doesn't even compare with an infant's learning capabilities

Intelligence and consciousness: The brain is highly intelligent and conscious, whereas the computer shows lack of intelligence

Evolution: The brains have been evolving for tens of millions of years, computers have been evolving for decades.

Spatial Brain Scales

1 m

CNS

10 cm

Systems

1 cm

Maps

1 mm

Networks

100 μm

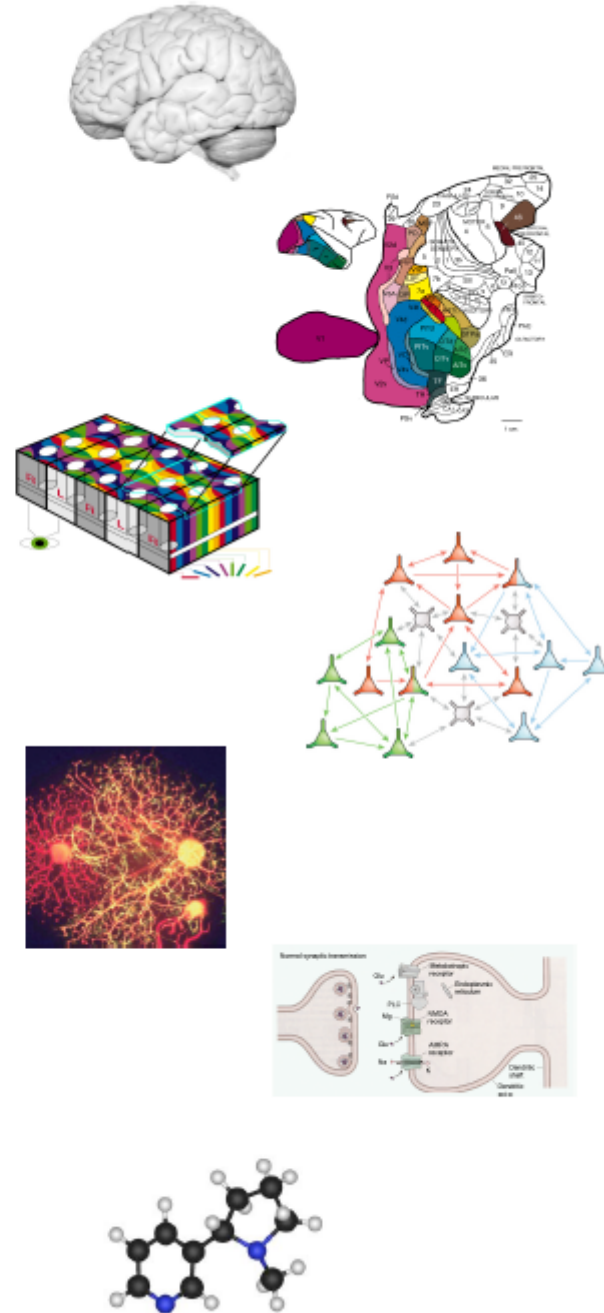
Neurons

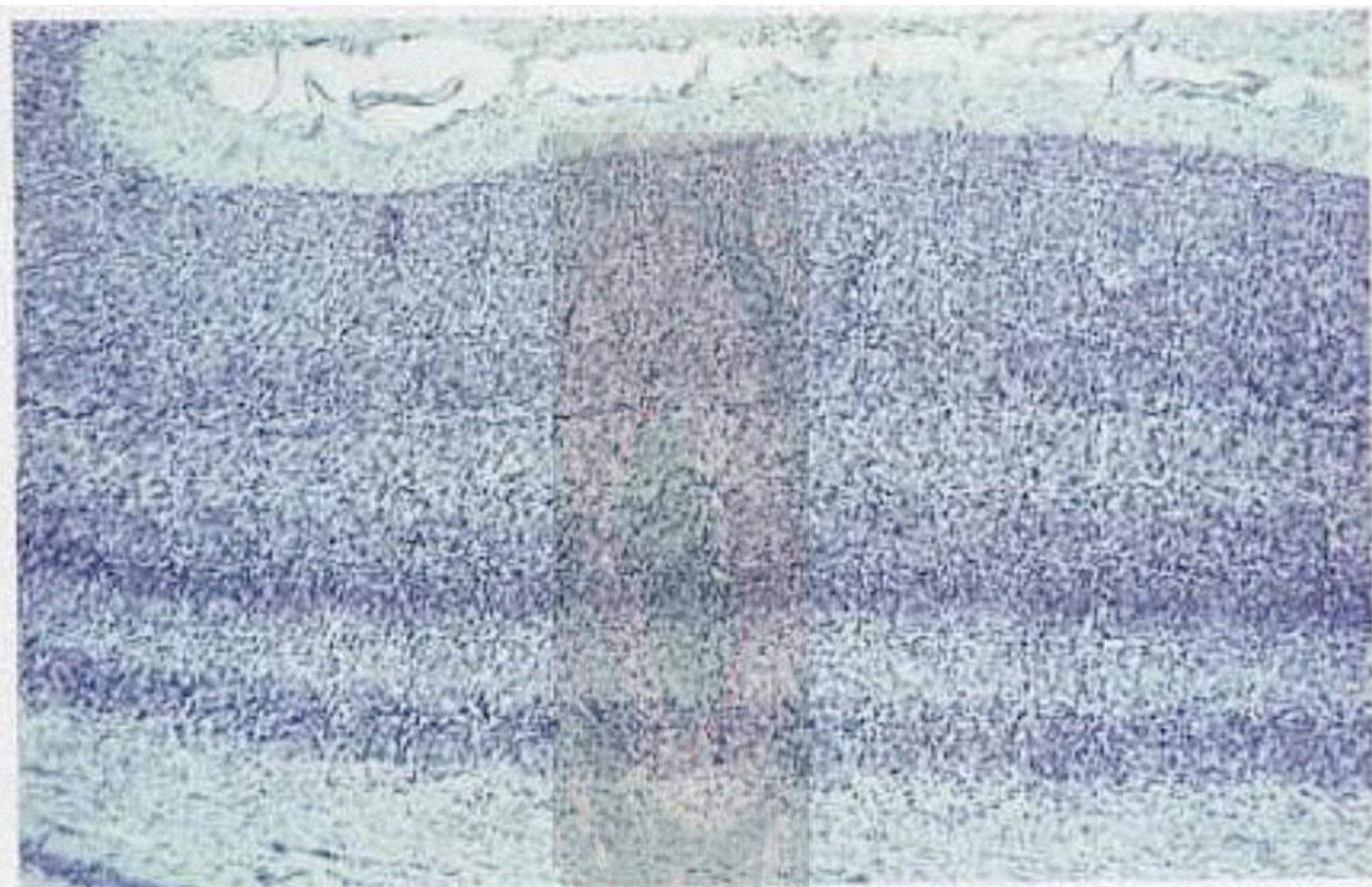
1 μm

Synapses

1 nm

Molecules





1

2

3

4A

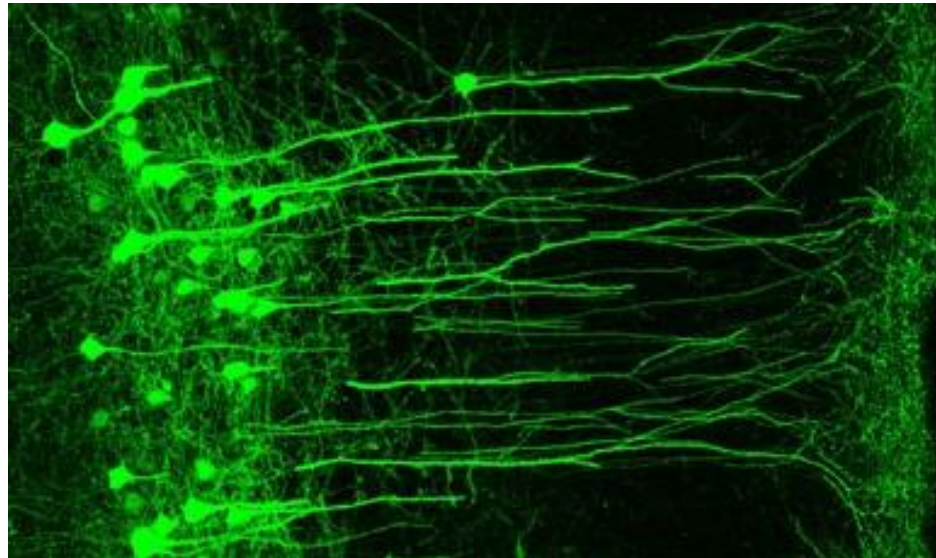
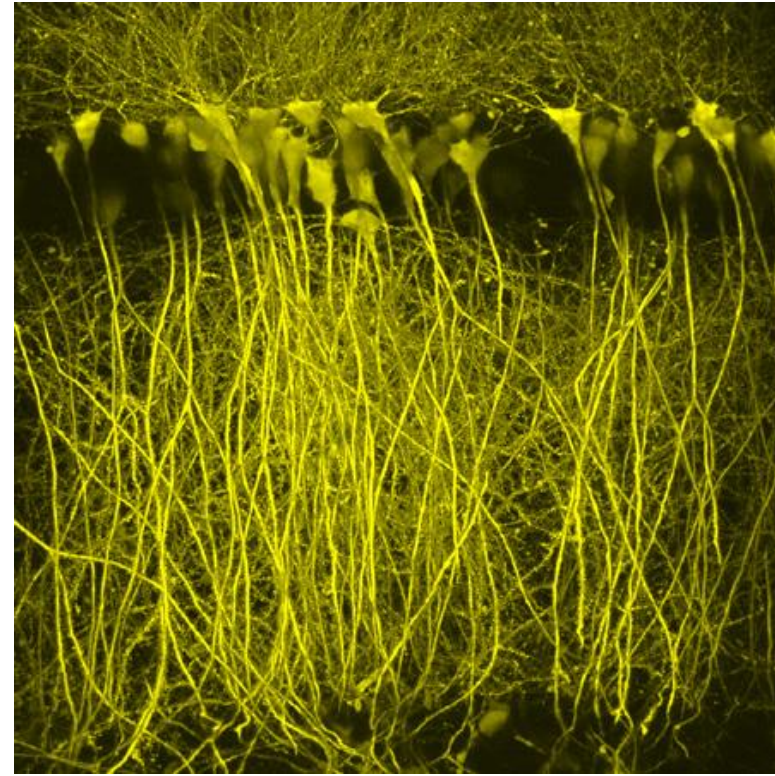
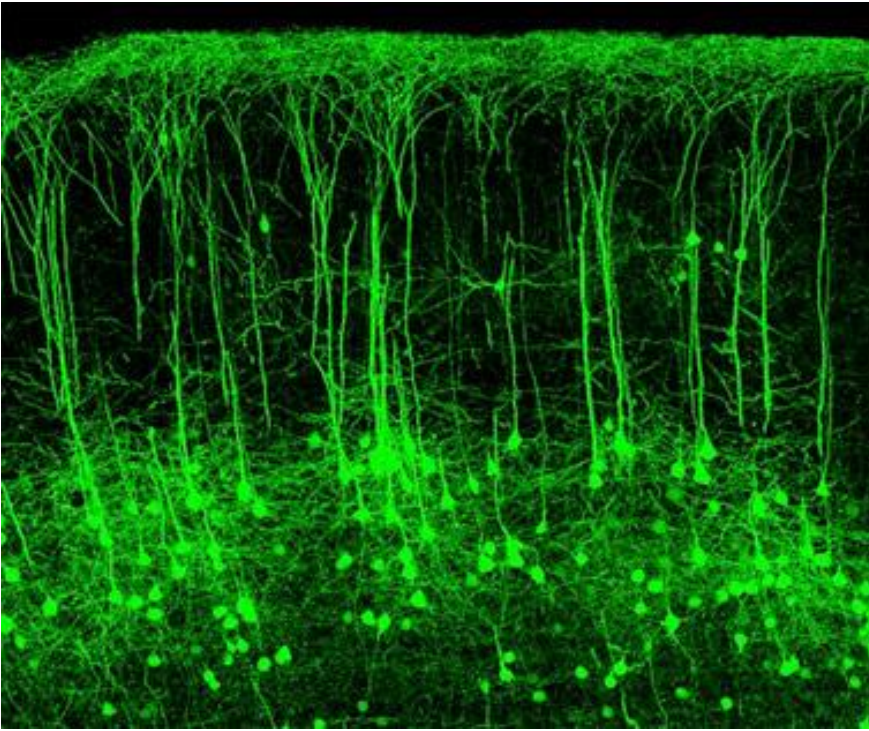
4B

4C

5

6

1 mm



Neural
Networks

What are Neural Nets used for?

There are two basic goals for neural network research:

Brain modelling : The *scientific* goal of building models of how real brains work.

This can potentially help us understand the nature of human intelligence, formulate better teaching strategies, or better remedial actions for brain damaged patients.

Artificial System Building : The *engineering* goal of building efficient systems for real world applications. This may make machines more powerful, relieve humans of tedious tasks, and may even improve upon human performance.

Not competing goals. Often exactly the same neural networks.

There are fundamental differences though,

e.g. the need for biological plausibility in brain modelling,

versus the need for computational efficiency in artificial system building.

Applications of Neural Nets

Neuroscience

Aid our understanding of how the brain works,
how behaviour emerges from the interaction of neurons.

Design of neural prosthetics – visual aids, robotic arms

Communication – brain damaged patients

Clinical marker – diagnosis and treatment of brain disorders

Applications in other fields

Financial modelling – predicting the stock market

Time series prediction – climate, weather, seizures

Computer games – intelligent agents, chess, backgammon

Robotics – autonomous adaptable robots

Pattern recognition – speech recognition, seismic activity, sonar signals

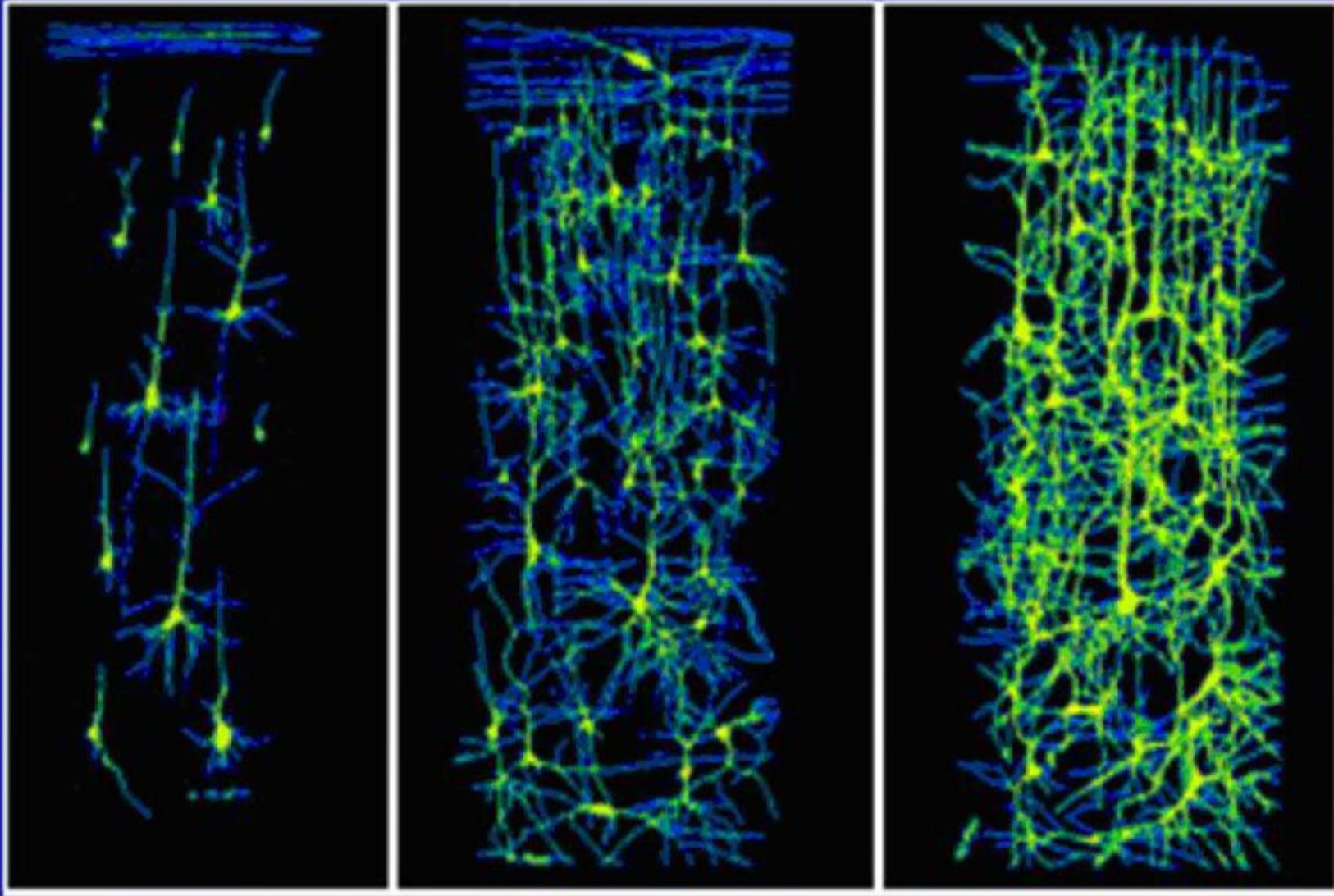
Data analysis – data compression, data mining

Bioinformatics – DNA sequencing, alignment

- **How do the neural connections form?**

- A minority of these connections comes with birth. But the majority are formed afterwards. Most of them occur in the first 2 years of life. During this period, $\sim 10^6$ connections are formed per second! This is called **hard wiring of the brain**. Development continues at later ages but slows down. During this period, connections may be made, later deleted, strengths of synapses modified etc. This is called **LEARNING**.

Synapse Formation



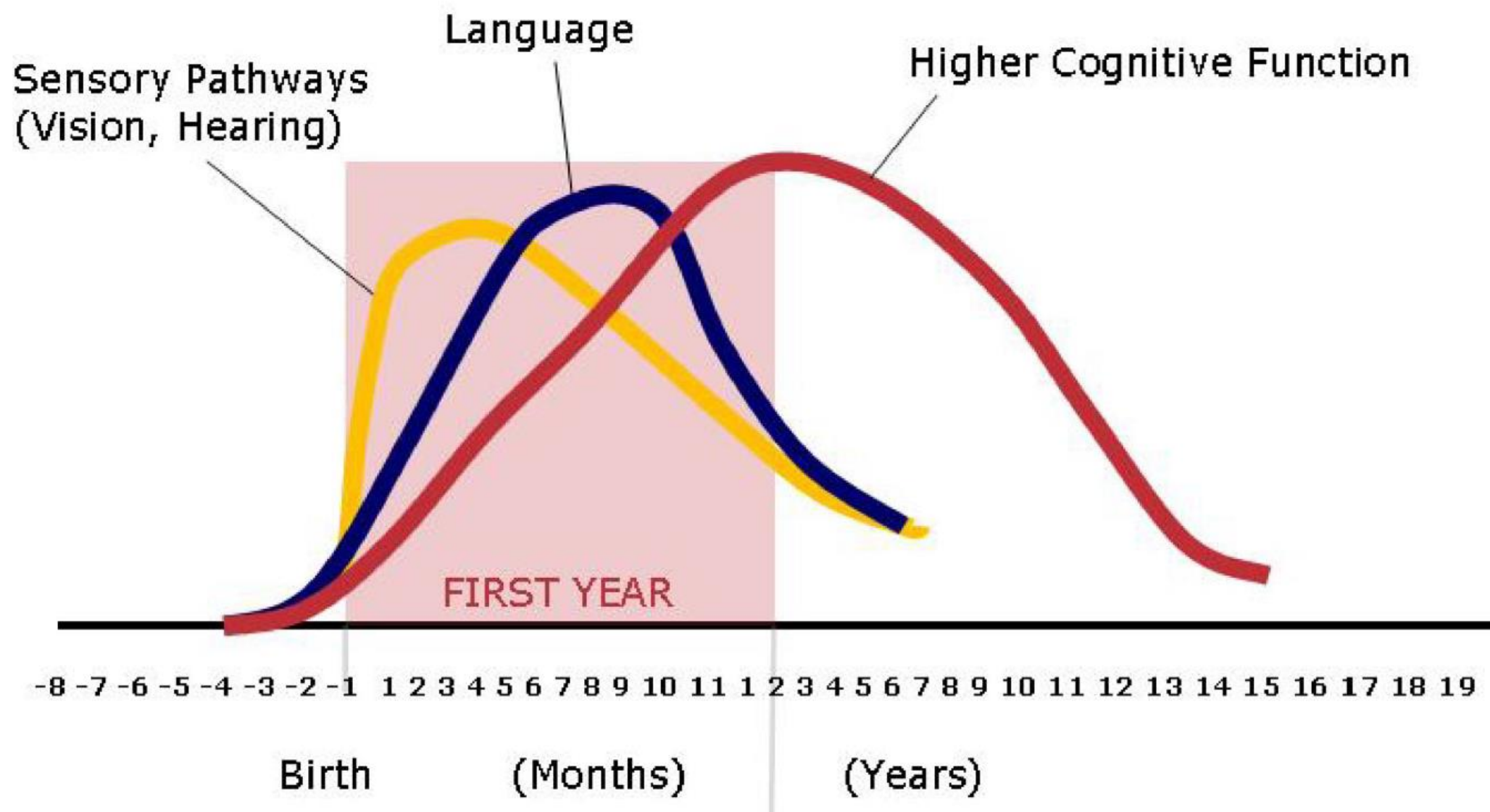
newborn

1 year old

2 year old

Human Brain Development

Synapse Formation Dependent on Early Experiences



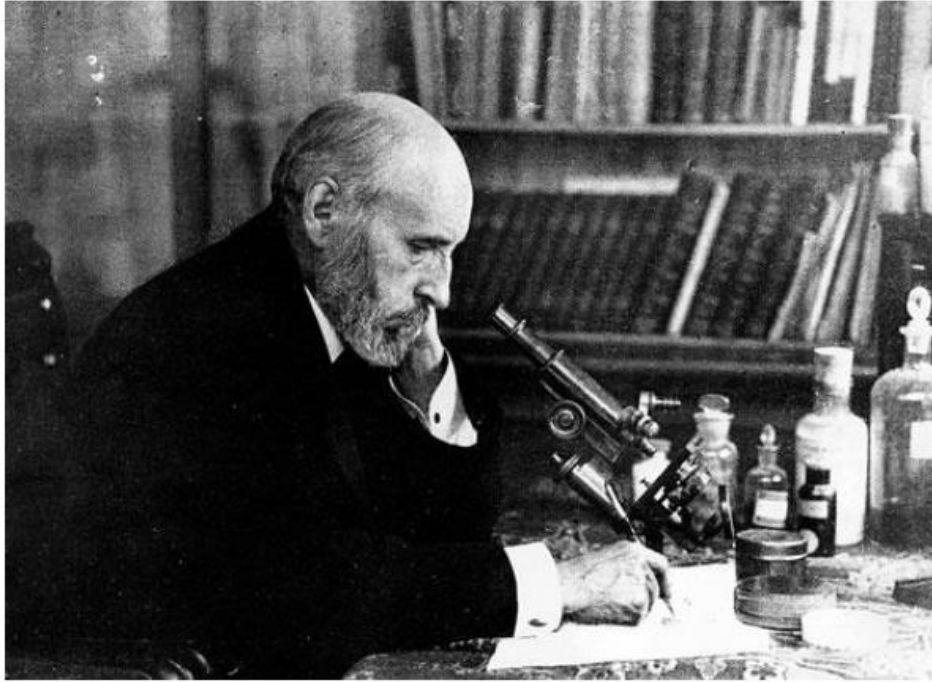
- **How does the brain decide on connections ?** → We do not know.

We concentrate on some form of optimization. Mainly, because of engineering/physical background this is the first think which comes to our mind.

- **Is the structure genetically coded ?** → Some part may be. For example, ability to see is inborn. But speech is not. One has to learn how to speak with the help of the environment. Even some properties related to vision (e.g. pattern recognition) must be learned. → Appropriate synaptic connections must be formed. → Kitten experiment..

A Brief History of the Field

- 1943** McCulloch and Pitts proposed the McCulloch-Pitts neuron model
- 1949** Hebb published his book *The Organization of Behavior*, in which the Hebbian learning rule was proposed.
- 1958** Rosenblatt introduced the simple single layer networks now called Perceptrons.
- 1969** Minsky and Papert's book *Perceptrons* demonstrated the limitation of single layer perceptrons, and almost the whole field went into hibernation.
- 1982** Hopfield published a series of papers on Hopfield networks.
- 1982** Kohonen developed the Self-Organising Maps that now bear his name.
- 1986** The Back-Propagation learning algorithm for Multi-Layer Perceptrons was re-discovered and the whole field took off again.
- 1990s** The sub-field of Radial Basis Function Networks was developed.
- 2000s** The power of Neural Network Ensembles, Support Vector Machines, Bayesian Techniques, Simulated Evolution, and Deep Learning becomes apparent.



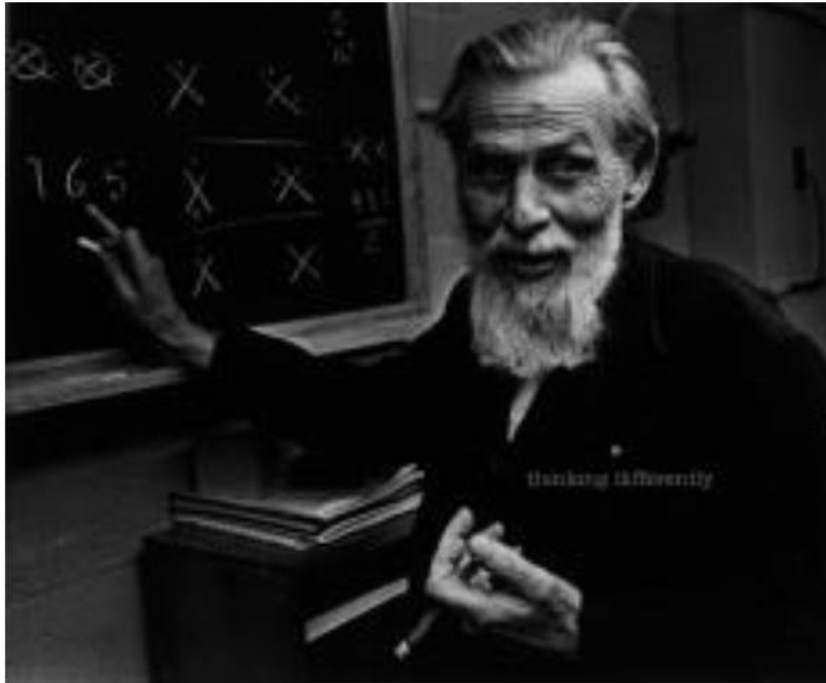
Santiago Ramón y Cajal



Camillo Golgi

Nobel Laureate (1906) ; Pathology - Neuroscience

Nervous system-synaptic gap-neuron models



Warren Sturgis McCulloch

Neurophysiology- Cybernetics

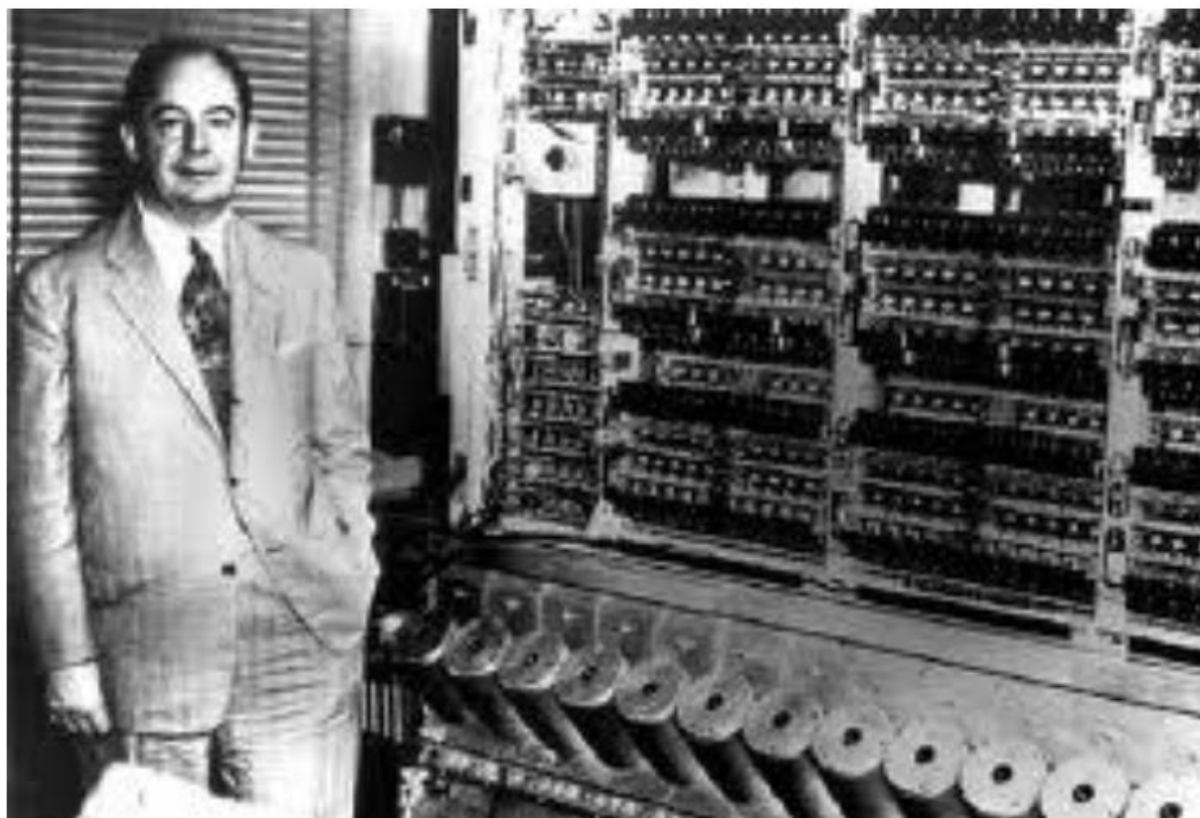


Walter Pitts

Logic – Cognitive Psychology

Basic arithmetic-logic operations

(1940~)

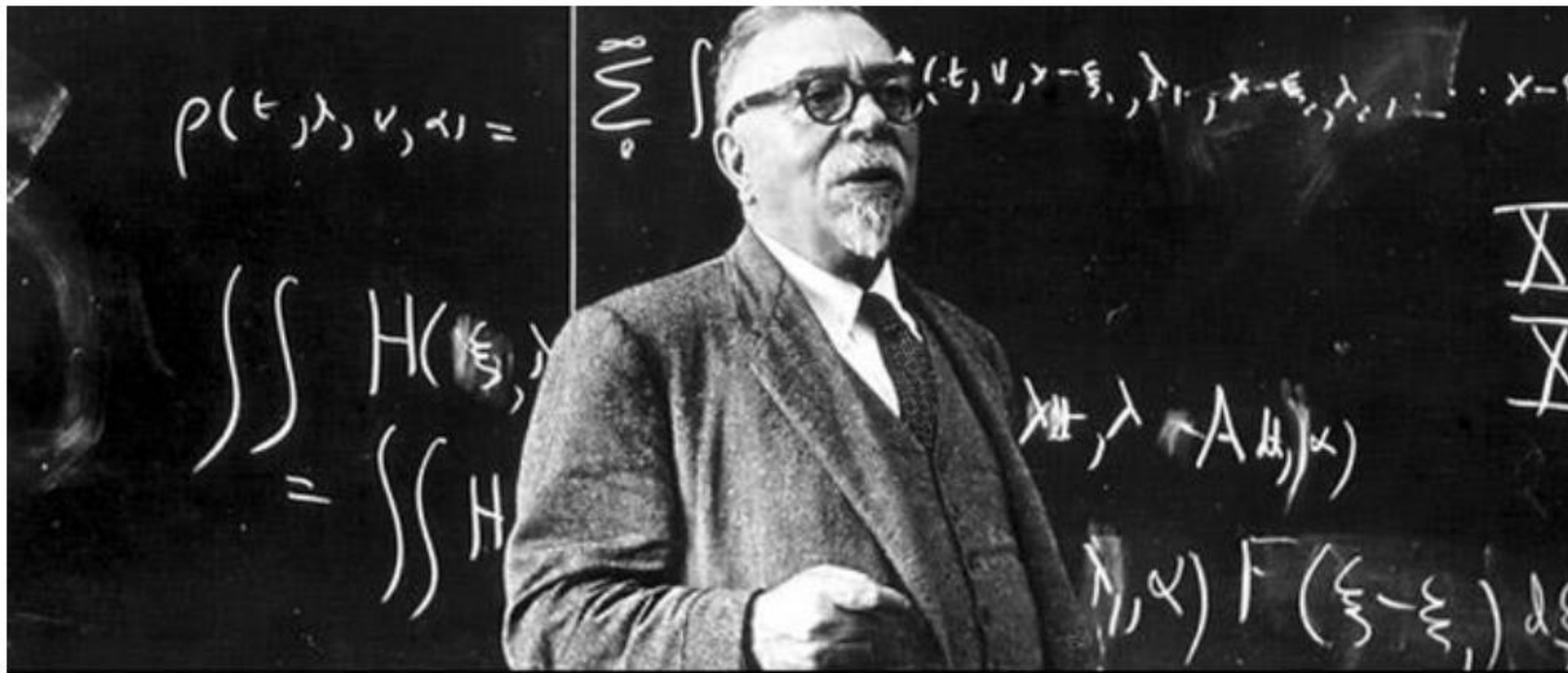


John von Neumann

Mathematician

First computer - brain vs. computers

(1950~)



Norbert Wiener

Mathematician

Cybernetics- statistical learning theory

(1950~)



Donald O. Hebb

Neuropsychology

Organization of behaviour - learning (1949)



Marvin Minsky

Cognitive Science – Artificial Intelligence

PhD in Neural Networks

(1954)



Frank Rosenblatt

Psychology – Computer Science

(1958)



Bernard Widrow



Ted Hoff

Electrical Engineer

Adaptive learning rule

(1960)



Stephen Grossberg



Gail Carpenter

Cognitive-Neuroscience-Biomedical Engineering- Mathematics

Adaptive learning – resonance theory (1980)



Teuvo Kohonen

Artificial Neural Networks – Artificial Intelligence

Self organization – Kohonen network (1982)



John Hopfield

Biology – Chemistry - Physics

Recurrent neural networks - memory (1982)



David Rumelhart

Psychology – Cognitive Science – Mathematical Psychology

Learning by backpropagation (1986)

Learning representations by back-propagating errors

David E. Rumelhart*, Geoffrey E. Hinton†
& Ronald J. Williams*

* Institute for Cognitive Science, C-015, University of California,
San Diego, La Jolla, California 92093, USA

† Department of Computer Science, Carnegie-Mellon University,
Pittsburgh, Philadelphia 15213, USA

We describe a new learning procedure, back-propagation, for networks of neurone-like units. The procedure repeatedly adjusts the weights of the connections in the network so as to minimize a measure of the difference between the actual output vector of the net and the desired output vector. As a result of the weight adjustments, internal 'hidden' units which are not part of the input or output come to represent important features of the task domain, and the regularities in the task are captured by the interactions of these units. The ability to create useful new features distinguishes back-propagation from earlier, simpler methods such as the perceptron-convergence procedure¹.

There have been many attempts to design self-organizing neural networks. The aim is to find a powerful synaptic modification rule that will allow an arbitrarily connected neural network to develop an internal structure that is appropriate for a particular task domain. The task is specified by giving the desired state vector of the output units for each state vector of the input units. If the input units are directly connected to the output units it is relatively easy to find learning rules that iteratively adjust the relative strengths of the connections so as to progressively reduce the difference between the actual and desired output vectors². Learning becomes more interesting but

more difficult when we introduce hidden units whose actual or desired states are not specified by the task. (In perceptrons, there are 'feature analysers' between the input and output that are not true hidden units because their input connections are fixed by hand, so their states are completely determined by the input vector: they do not learn representations.) The learning procedure must decide under what circumstances the hidden units should be active in order to help achieve the desired input-output behaviour. This amounts to deciding what these units should represent. We demonstrate that a general purpose and relatively simple procedure is powerful enough to construct appropriate internal representations.

The simplest form of the learning procedure is for layered networks which have a layer of input units at the bottom; any number of intermediate layers; and a layer of output units at the top. Connections within a layer or from higher to lower layers are forbidden, but connections can skip intermediate layers. An input vector is presented to the network by setting the states of the input units. Then the states of the units in each layer are determined by applying equations (1) and (2) to the connections coming from lower layers. All units within a layer have their states set in parallel, but different layers have their states set sequentially, starting at the bottom and working upwards until the states of the output units are determined.

The total input, x_j , to unit j is a linear function of the outputs, y_i , of the units that are connected to j and of the weights, w_{ji} , on these connections

$$x_j = \sum_i y_i w_{ji} \quad (1)$$

Units can be given biases by introducing an extra input to each unit which always has a value of 1. The weight on this extra input is called the bias and is equivalent to a threshold of the opposite sign. It can be treated just like the other weights.

A unit has a real-valued output, y_j , which is a non-linear function of its total input

$$y_j = \frac{1}{1 + \exp(-x_j)} \quad (2)$$

The Godfather of Deep Learning



Vin Busquet · Follow

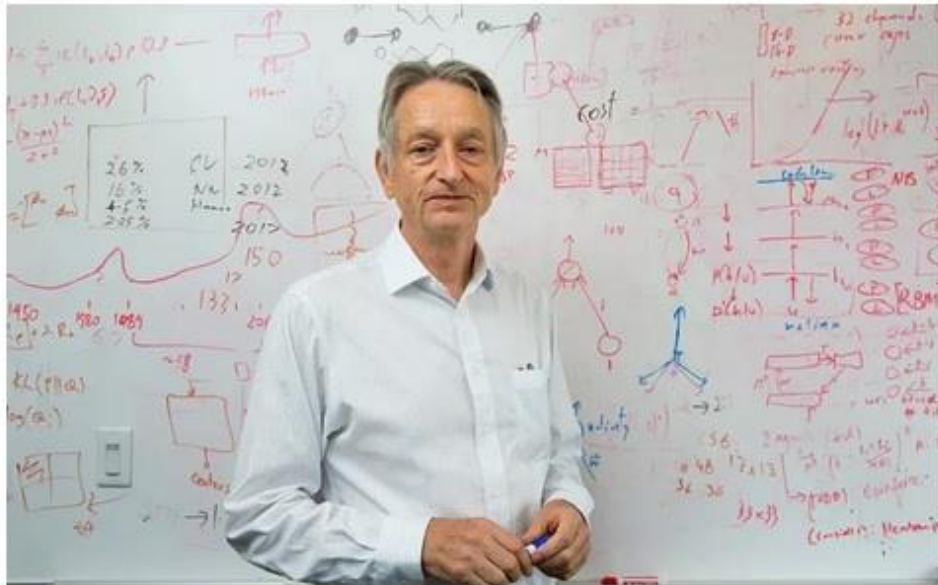
4 min read · Jan 9, 2020



22



Geoffrey Hinton is one of the geniuses who helped create modern AI



Professor Geoffrey Hinton CREDIT: [JULIAN SIMMONDS](#)

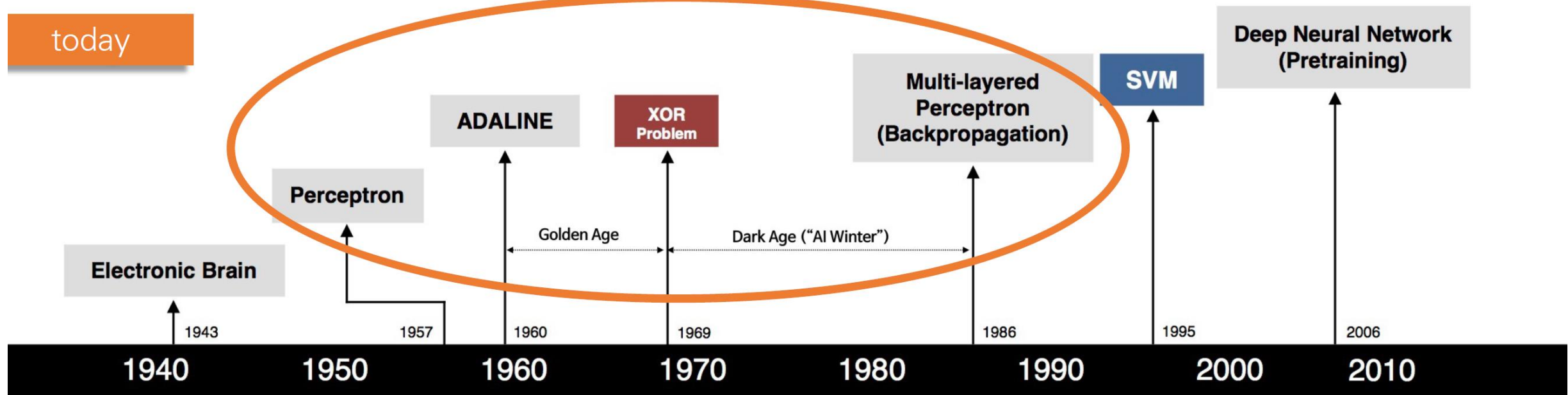
What is Deep Learning?

Recently Deep Learning has become the hottest subject in the field of artificial intelligence, thanks in particular to breakthroughs in image and language recognition in recent years that have approached or surpassed human levels of comprehension.

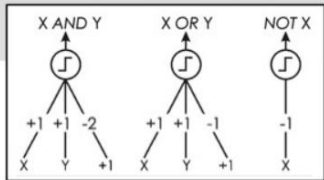
It is basically a subfield of machine learning algorithms inspired by the structure and function of the brain called **artificial neural networks**. Deep neural networks uses multiples layers or artificial neurons to progressively extract higher level features from the raw input.

<https://medium.com/@vibusquet/the-godfather-of-deep-learning-2f8d38ed8a79>

A Brief History of Neural Networks



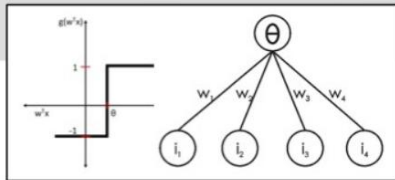
S. McCulloch – W. Pitts



- Adjustable Weights
- Weights are not Learned



F. Rosenblatt



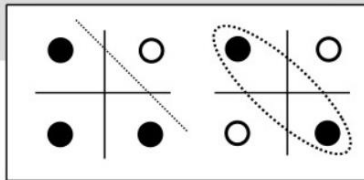
- Learnable Weights and Threshold



B. Widrow – M. Hoff



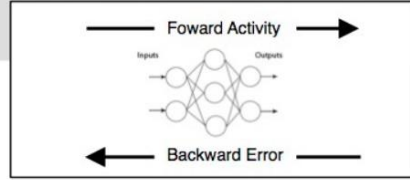
M. Minsky – S. Papert



- XOR Problem



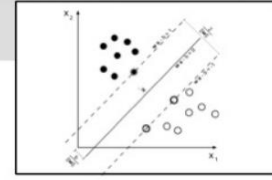
D. Rumelhart – G. Hinton – R. Williams



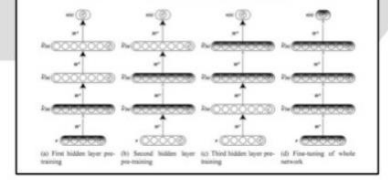
- Solution to nonlinearly separable problems
- Big computation, local optima and overfitting
- Limitations of learning prior knowledge
- Kernel function: Human Intervention



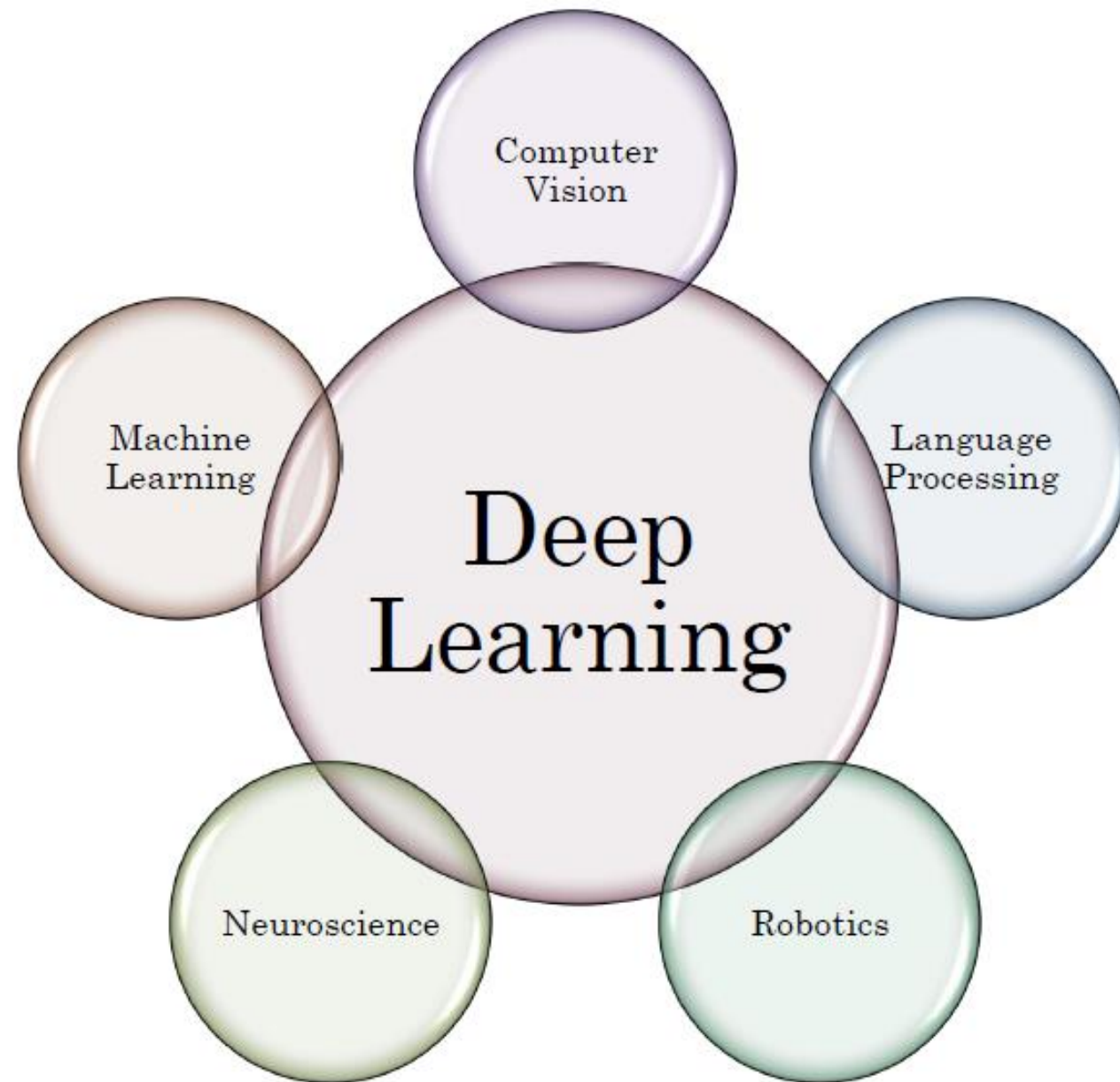
V. Vapnik – C. Cortes



G. Hinton – S. Ruslan

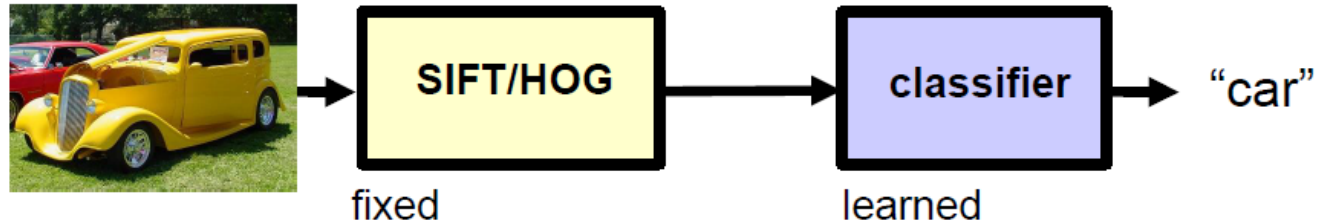


- Hierarchical feature Learning

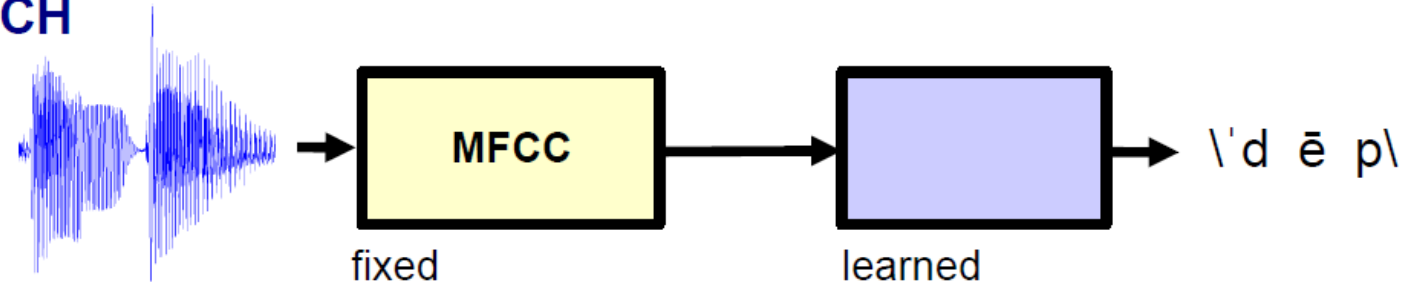


Traditional Machine Learning

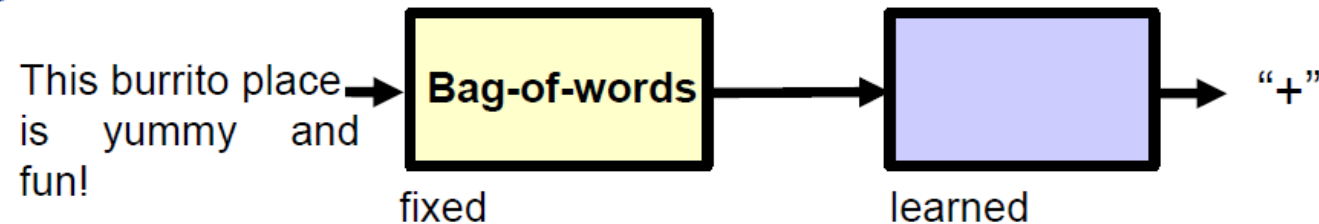
VISION



SPEECH



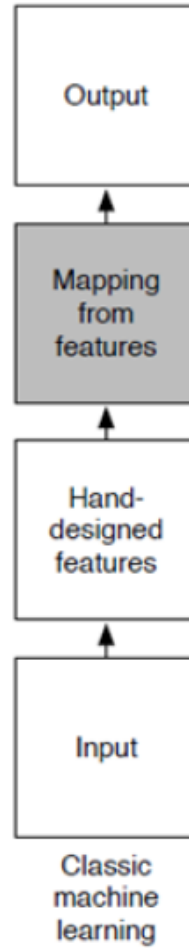
NLP



Slide Credit: Marc'Aurelio Ranzato, Yann LeCun

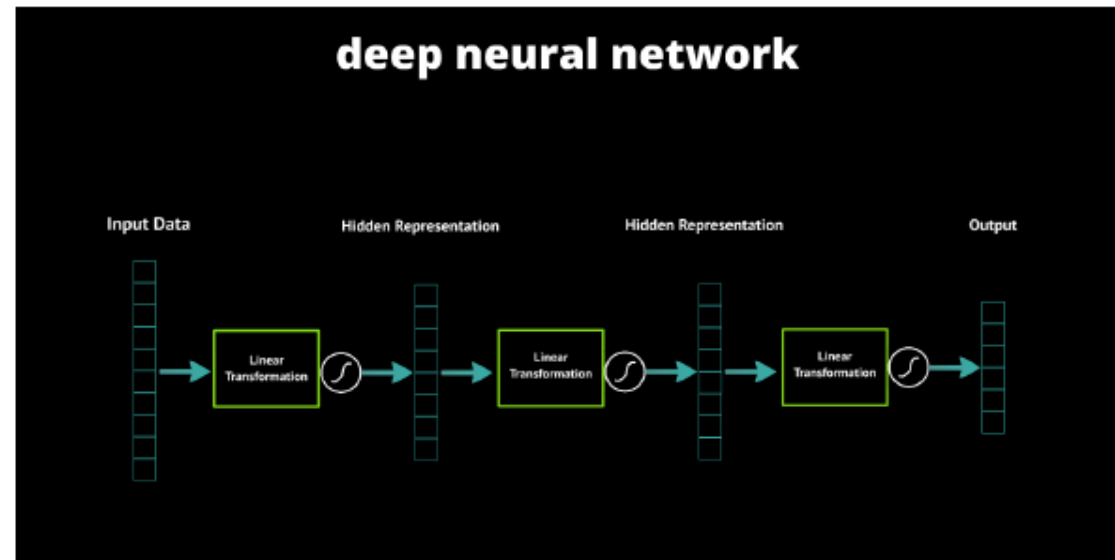
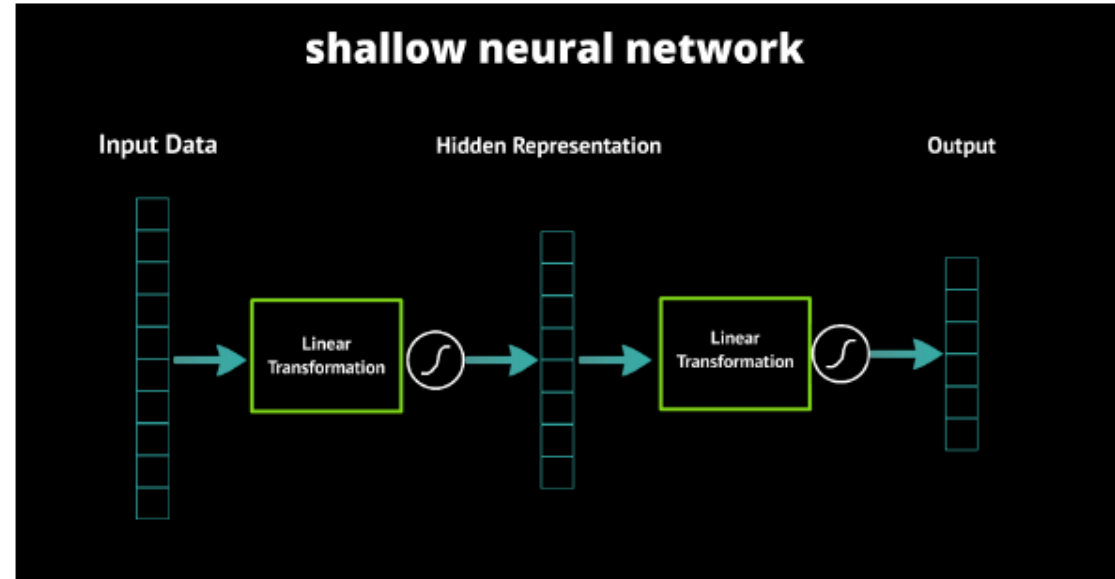
Ac

Deep learning

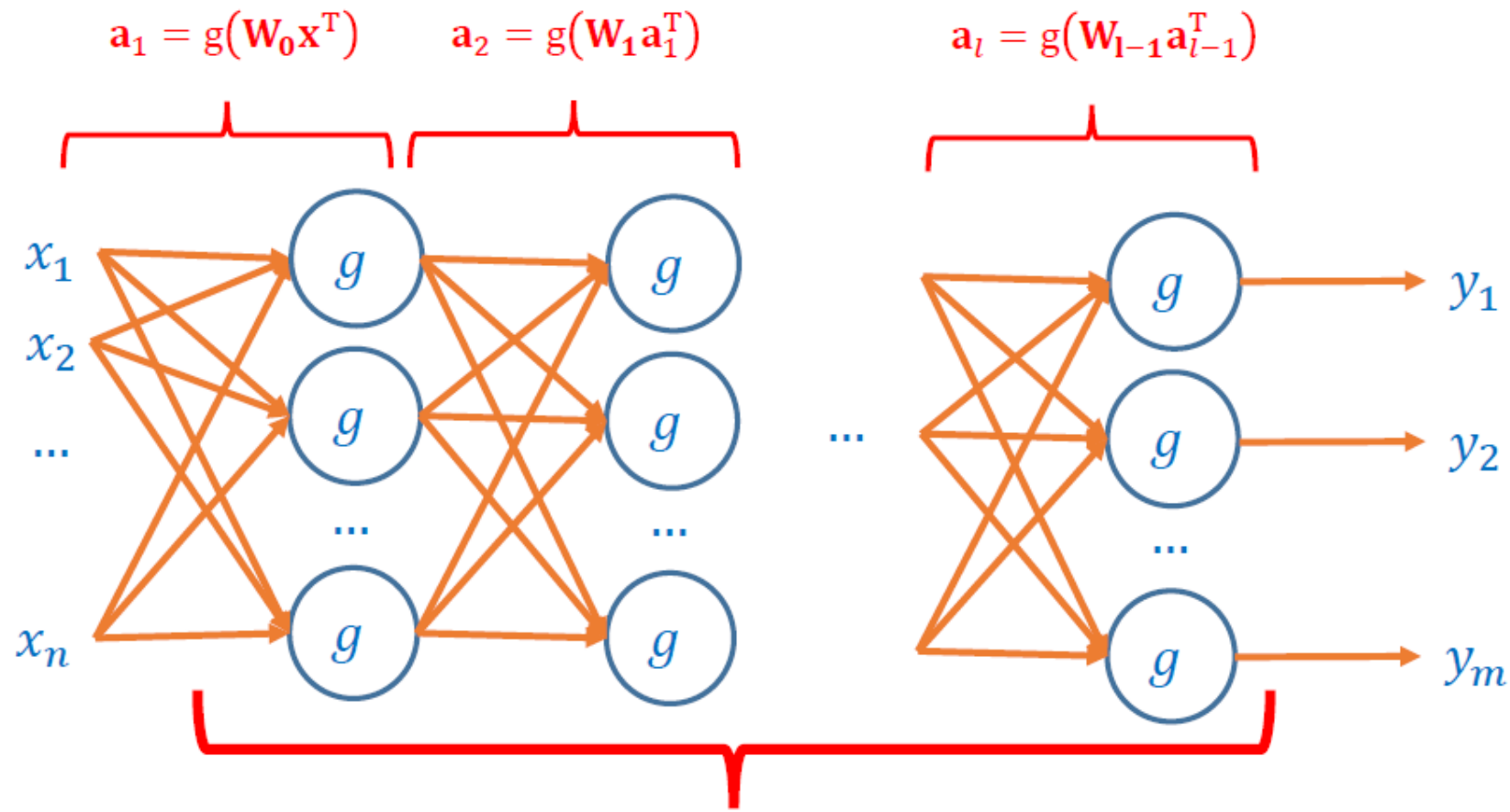


But, what exactly is deep learning?

Deep neural networks have more than one hidden layer. It really is that simple.

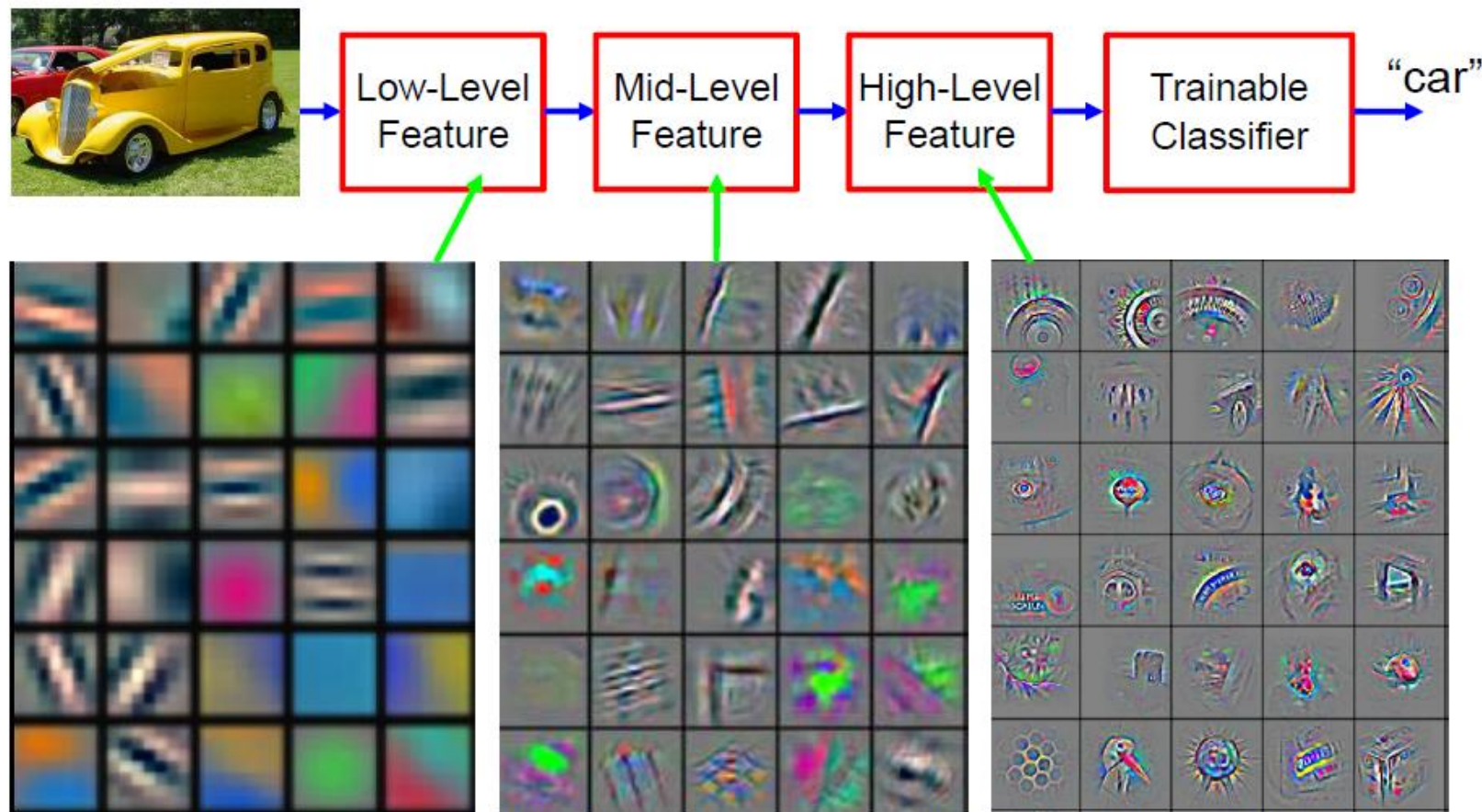


Main idea in DL



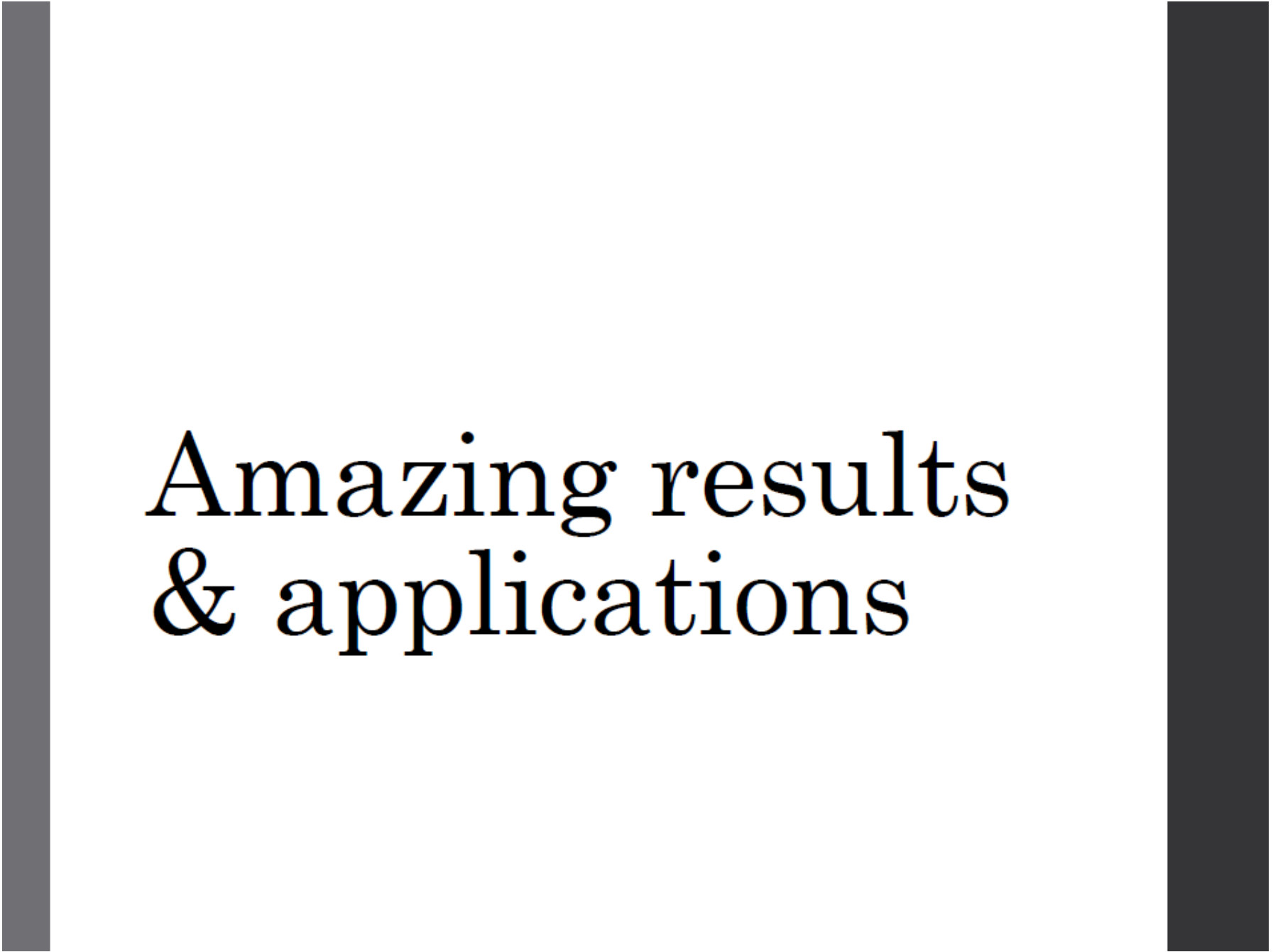
$$f() = g \left(g \left(g \left(g (\dots) \right) \right) \right)$$

Deep Learning = Hierarchical Compositionality



Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

Slide Credit: Marc'Aurelio Ranzato, Yann LeCun



Amazing results
& applications

ImageNet Object Recognition Challenge

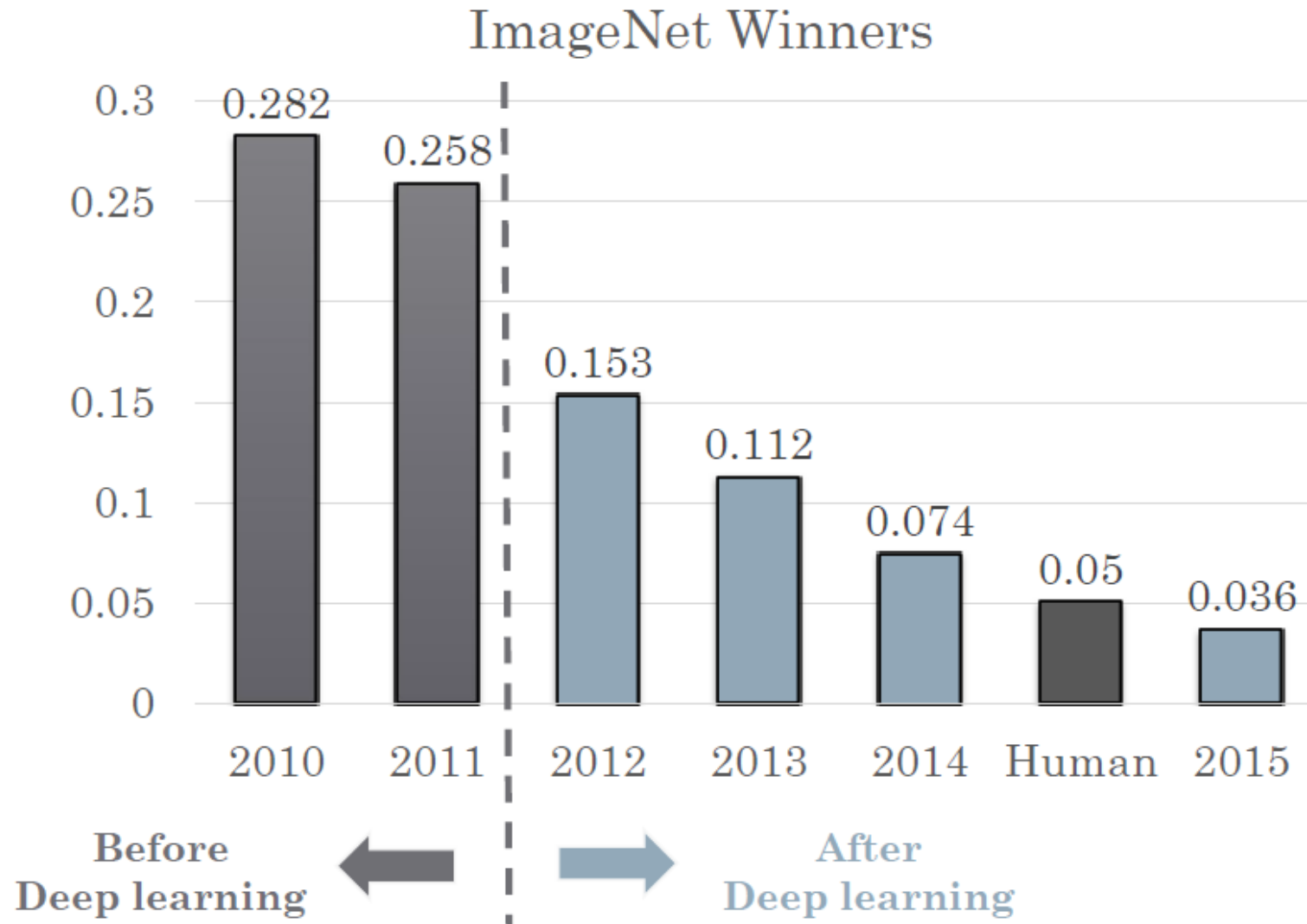
(<http://image-net.org/challenges/LSVRC/>)



Alex Krizhevsky, Ilya Sutskever, Geoffrey E. Hinton,
ImageNet Classification with Deep Convolutional Neural
Networks, NIPS, 2012.

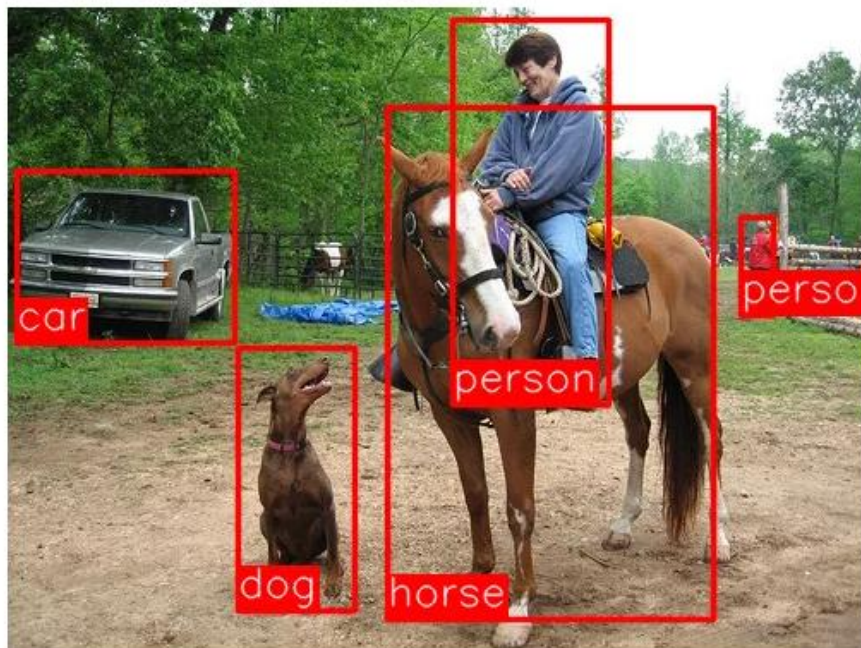
<http://www.rctn.org/bruno/public/papers/deep-learning/2012%20Krizhevsky.pdf>

Object Recognition

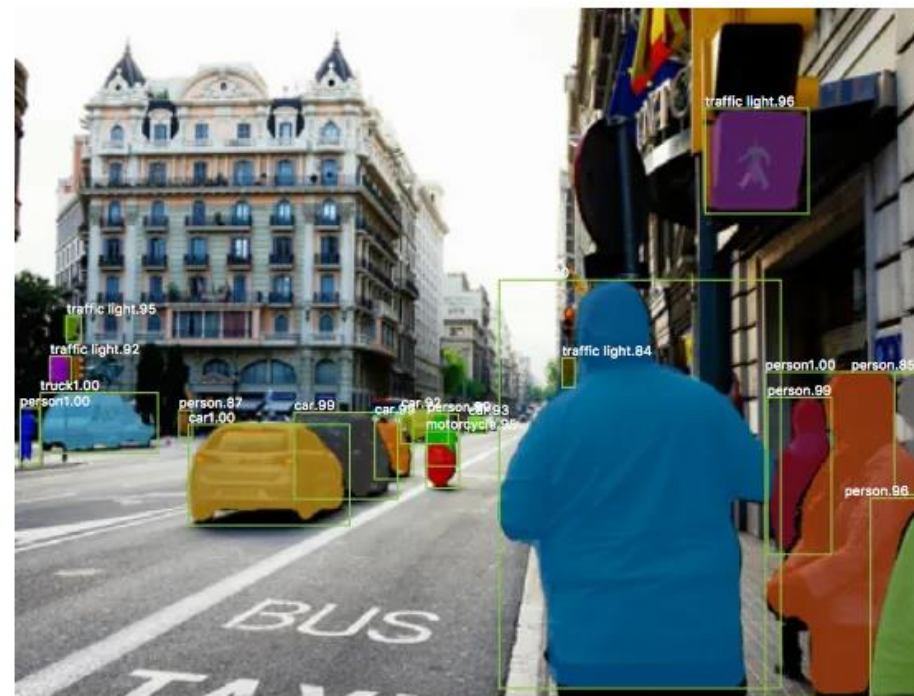


<https://en.wikipedia.org/wiki/ImageNet>

2014: R-CNN - An Early Application of CNNs to Object Detection



Object detection algorithms such as R-CNN take in an image and identify the locations and classifications of the main objects in the image. Source: <https://arxiv.org/abs/1311.2524>.



In image segmentation, our goal is to classify the different objects in the image, and identify their boundaries. Source: Mask R-CNN paper.

<https://blog.athelas.com/a-brief-history-of-cnns-in-image-segmentation-from-r-cnn-to-mask-r-cnn-34ea83205de4>

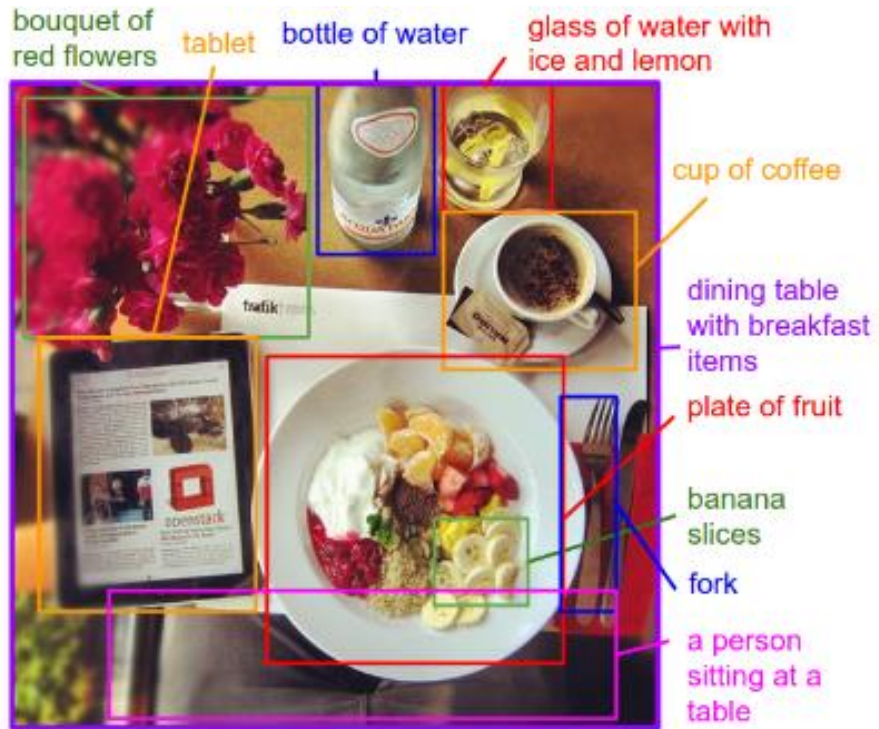


Figure 1. Motivation/Concept Figure: Our model treats language as a rich label space and generates descriptions of image regions.



two young girls are playing with lego toy.

Karpathy & Fei-Fei, 2015

https://www.cv-foundation.org/openaccess/content_cvpr_2015/papers/Karpathy_Deep_Visual-Semantic_Alignments_2015_CVPR_paper.pdf



https://www.youtube.com/watch?v=_HCWBS6k8j0



(Google, 2017)

<https://www.youtube.com/watch?v=gn4nRCC9TwQ>

Text Generation

Proof. Omitted. □

Lemma 0.1. *Let $\mathcal{C}$ be a set of the construction.*

Let $\mathcal{C}$ be a gerber covering. Let $\mathcal{F}$ be a quasi-coherent sheaves of $\mathcal{O}$ -modules. We have to show that

$$\mathcal{O}_{\mathcal{O}_X} = \mathcal{O}_X(\mathcal{L})$$

Proof. This is an algebraic space with the composition of sheaves $\mathcal{F}$ on $X_{\text{étale}}$ we have

$$\mathcal{O}_X(\mathcal{F}) = \{\text{morph}_1 \times_{\mathcal{O}_X} (\mathcal{G}, \mathcal{F})\}$$

where $\mathcal{G}$ defines an isomorphism $\mathcal{F} \rightarrow \mathcal{F}$ of $\mathcal{O}$ -modules. □

Lemma 0.2. *This is an integer $\mathbb{Z}$ is injective.*

Proof. See Spaces, Lemma ?? □

Lemma 0.3. *Let S be a scheme. Let X be a scheme and X is an affine open covering. Let $\mathcal{U} \subset \mathcal{X}$ be a canonical and locally of finite type. Let X be a scheme. Let X be a scheme which is equal to the formal complex.*

The following to the construction of the lemma follows.

Let X be a scheme. Let X be a scheme covering. Let

$$b : X \rightarrow Y' \rightarrow Y \rightarrow Y \rightarrow Y' \times_X Y \rightarrow X.$$

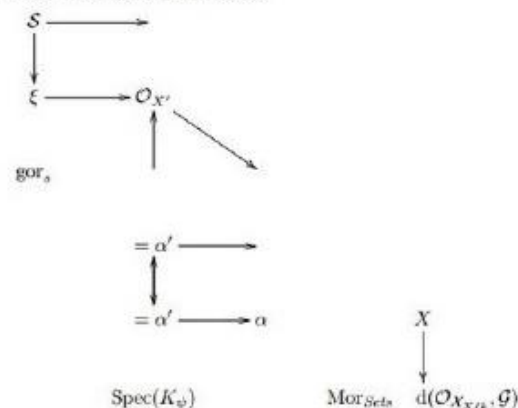
be a morphism of algebraic spaces over S and Y .

Proof. Let X be a nonzero scheme of X . Let X be an algebraic space. Let $\mathcal{F}$ be a quasi-coherent sheaf of $\mathcal{O}_X$ -modules. The following are equivalent

- (1) $\mathcal{F}$ is an algebraic space over S .
- (2) If X is an affine open covering.

Consider a common structure on X and X the functor $\mathcal{O}_X(U)$ which is locally of finite type. □

This since $\mathcal{F} \in \mathcal{F}$ and $x \in \mathcal{G}$ the diagram



is a limit. Then $\mathcal{G}$ is a finite type and assume S is a flat and $\mathcal{F}$ and $\mathcal{G}$ is a finite type f_* . This is of finite type diagrams, and

- the composition of $\mathcal{G}$ is a regular sequence,
- $\mathcal{O}_{X'}$ is a sheaf of rings.

□

Proof. We have see that $X = \text{Spec}(R)$ and $\mathcal{F}$ is a finite type representable by algebraic space. The property $\mathcal{F}$ is a finite morphism of algebraic stacks. Then the cohomology of X is an open neighbourhood of U . □

Proof. This is clear that $\mathcal{G}$ is a finite presentation, see Lemmas ??.

A reduced above we conclude that U is an open covering of $\mathcal{C}$. The functor $\mathcal{F}$ is a "field"

$$\mathcal{O}_{X,x} \rightarrow \mathcal{F}_x \rightarrow \mathcal{O}_{X,x}^{-1}(\mathcal{O}_{X_{\text{étale}}}) \rightarrow \mathcal{O}_{X,x}^{-1}(\mathcal{O}_{X_{\text{étale}}})$$

is an isomorphism of covering of $\mathcal{O}_{X_1}$. If $\mathcal{F}$ is the unique element of $\mathcal{F}$ such that X is an isomorphism.

The property $\mathcal{F}$ is a disjoint union of Proposition ?? and we can filtered set of presentations of a scheme $\mathcal{O}_X$ -algebra with $\mathcal{F}$ are opens of finite type over $\mathcal{S}$. If $\mathcal{F}$ is a scheme theoretic image points. □

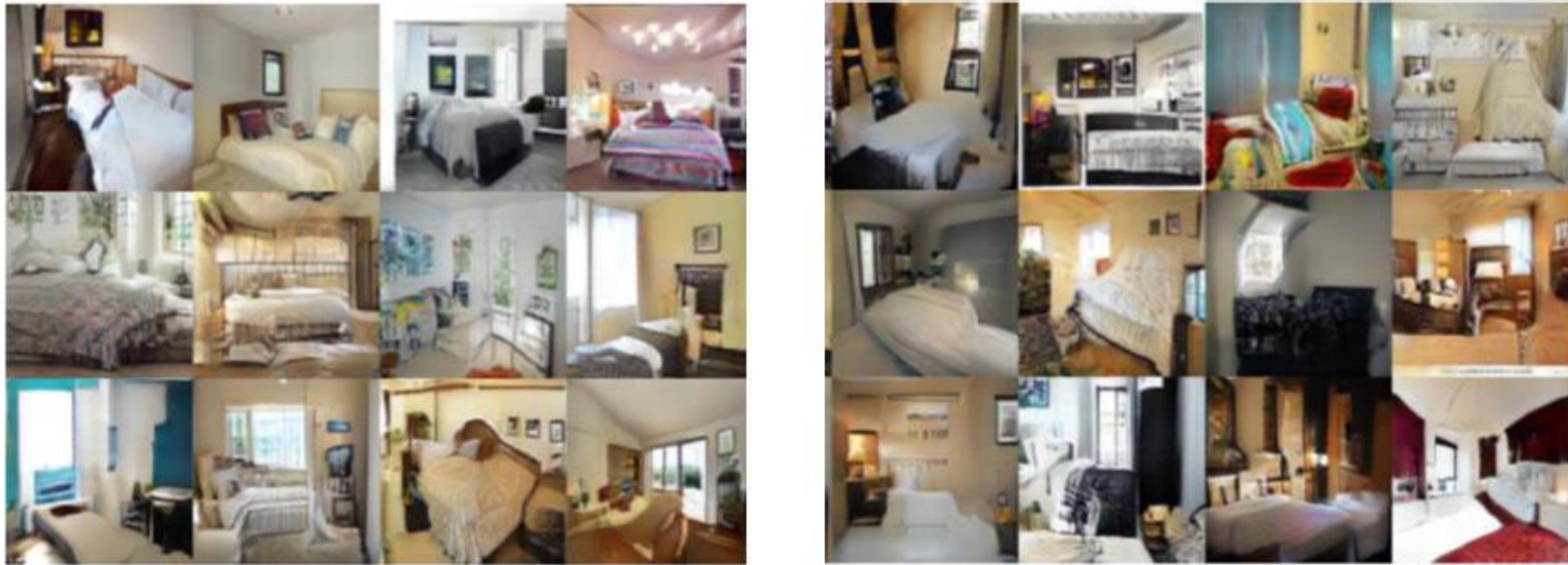
If $\mathcal{F}$ is a finite direct sum $\mathcal{O}_{X_1}$ is a closed immersion, see Lemma ?? . This is a sequence of $\mathcal{F}$ is a similar morphism.

<http://karpathy.github.io/2015/05/21/rnn-effectiveness/>

<http://karpathy.github.io/2015/05/21/rnn-effectiveness/>

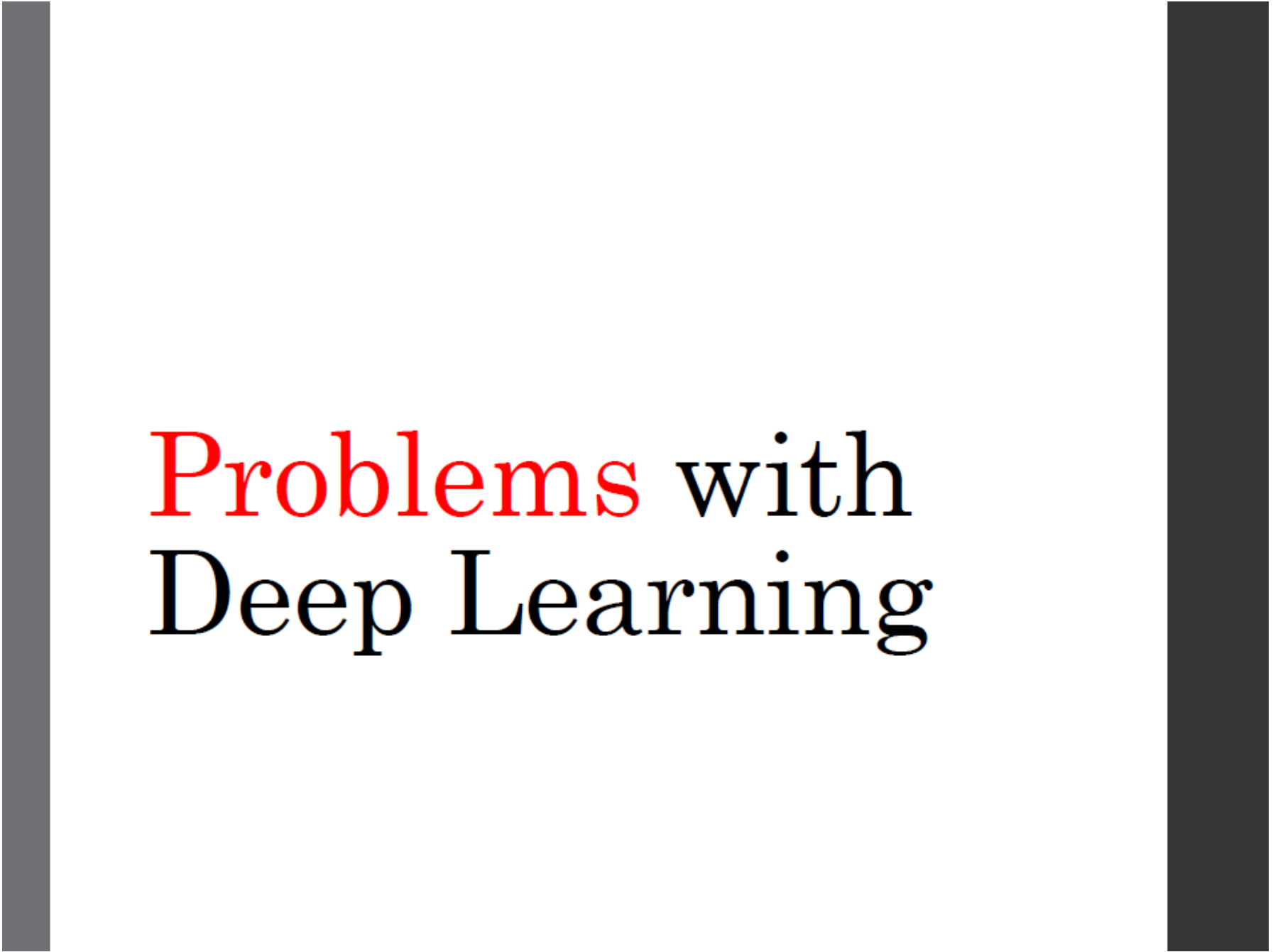
Image Generation

Which ones are real? The ones
on the left, or the right?



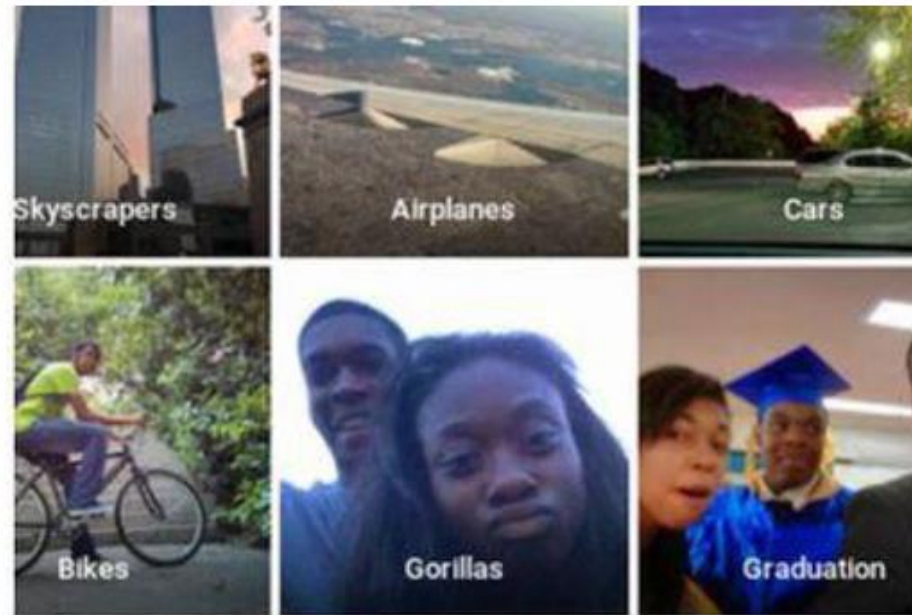
Xudong Mao, Qing Liy, Haoran Xiez, Raymond Y.K. Laux, and
Zhen Wang, Least Squares Generative Adversarial Networks,

https://openaccess.thecvf.com/content_ICCV_2017/papers/Mao_Least_Squares_Generative_ICCV_2017_paper.pdf



Problems with Deep Learning

It can make mistakes

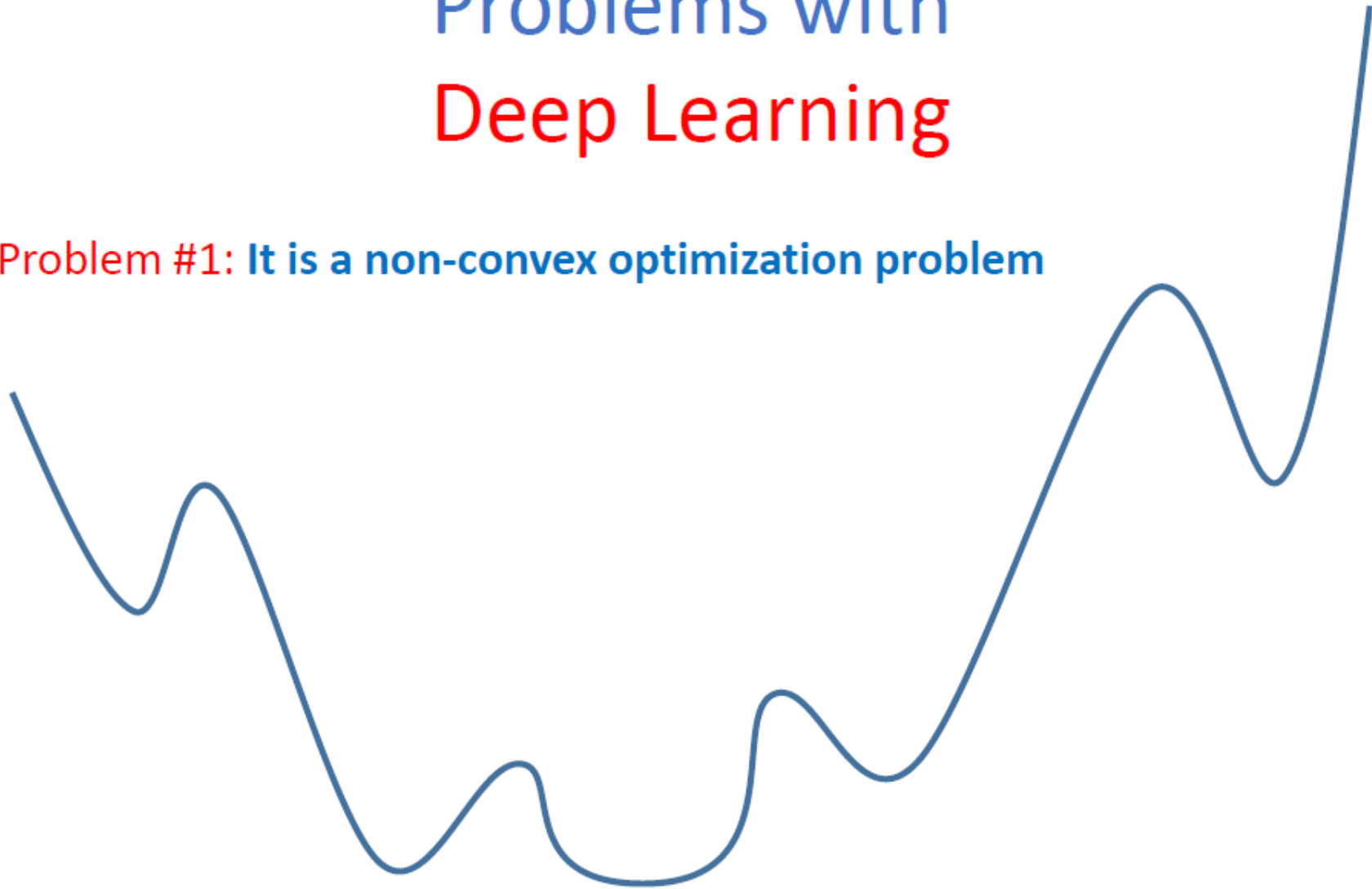


<http://blogs.wsj.com/digits/2015/07/01/google-mistakenly-tags-black-people-as-gorillas-showing-limits-of-algorithms/>

<https://www.bbc.com/news/technology-33347866>

Problems with Deep Learning

Problem #1: It is a non-convex optimization problem



- Problem#2: Hard to track down what's failing



Figure 5. A selection of evaluation results, grouped by human rating.



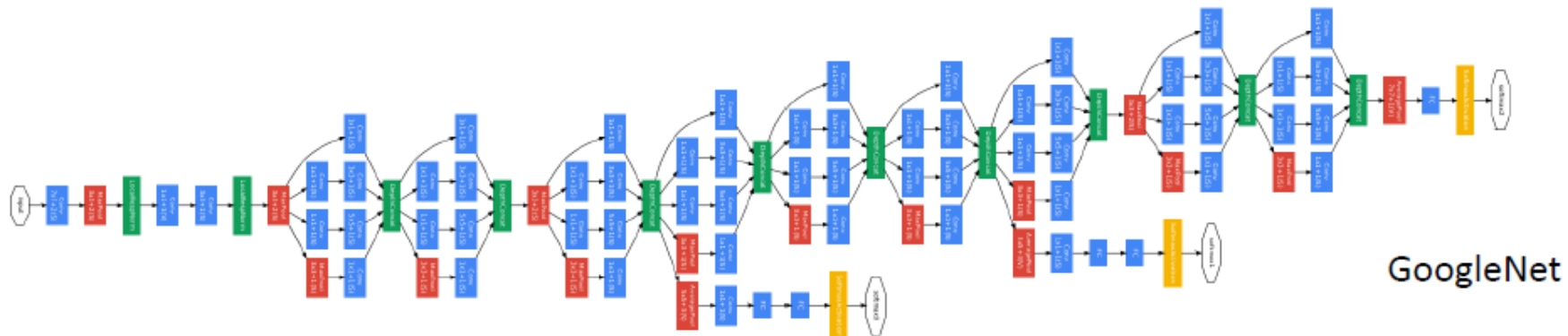
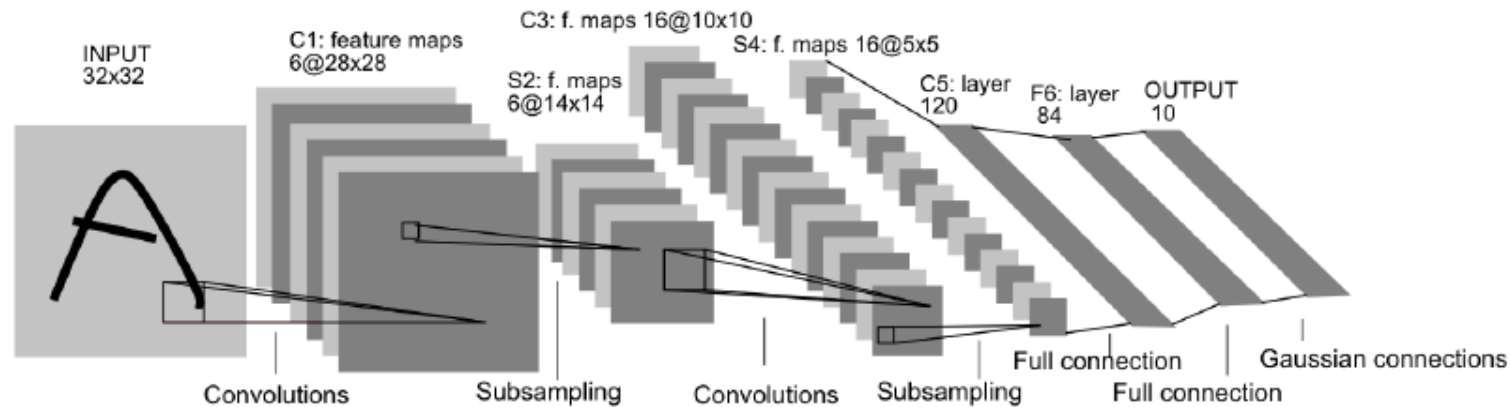
Figure 1. An illustrative example of our pipeline.

https://openaccess.thecvf.com/content_cvpr_2015/papers/Fang_From_Captions_to_2015_CVPR_paper.pdf

Problems with Deep Learning

- Problem#3: Lack of easy reproducibility
 - Direct consequence of stochasticity & non-convexity
- Standard response #1
 - It's getting much better
 - Standard toolkits/libraries/frameworks now available
 - Caffe, Theano, Torch
- Standard response #2
 - “Yes, but it often works!”

Problems with Deep Learning: Architecture Engineering



Problems with Deep Learning

Requires too much data

Requires too much computational power

Big players

Google, Microsoft, Facebook, Apple, Amazon, Intel, Nvidia

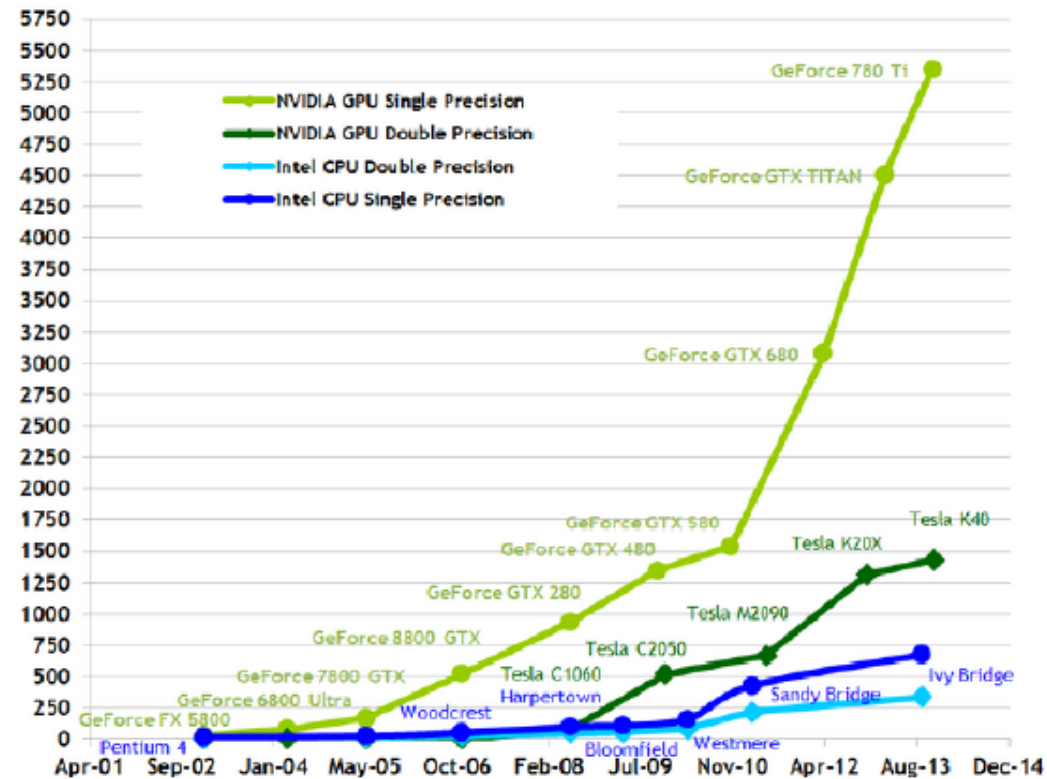
Why does it work now?

We have more computational power now.



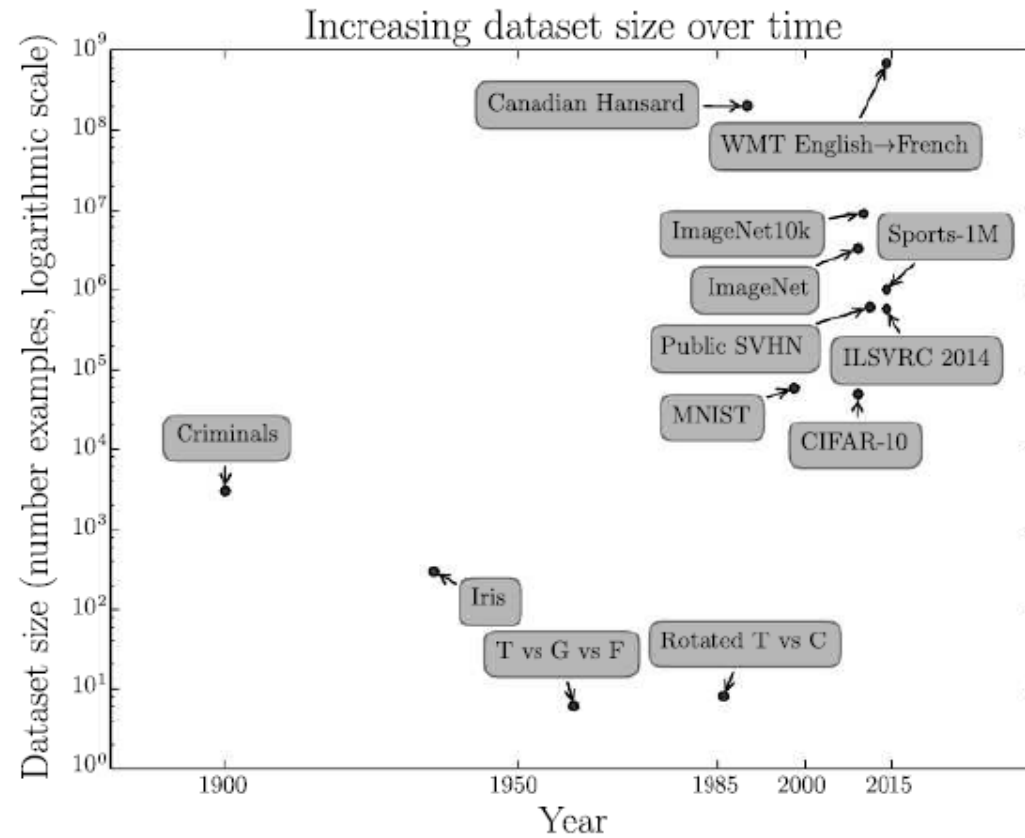
Mahmoud Ahmed
Hossam, High
Performance
Hyperspectral Image
Classification using
Graphics Processing
Units, 2015

Theoretical GFLOP/s

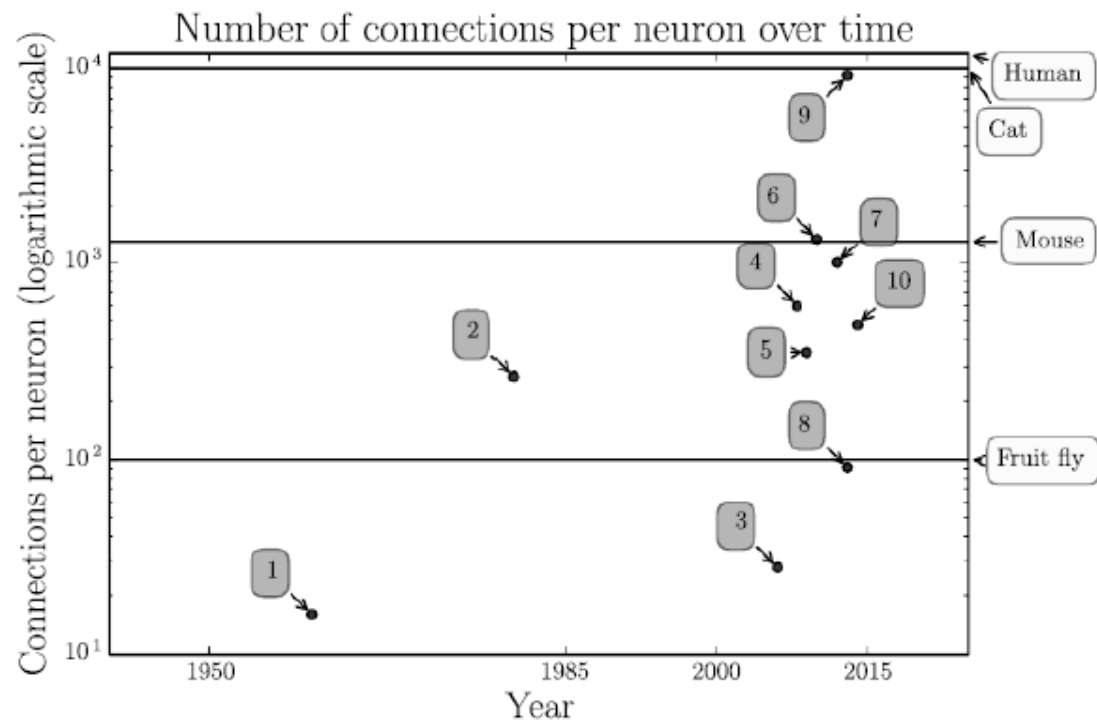


Why does it work now?

We have more data now.



More
connections
now



1. Adaptive Linear Element (Widrow and Hoff, 1960)
2. Neocognitron (Fukushima, 1980)
3. GPU-accelerated convolutional network (Chellapilla *et al.*, 2006)
4. Deep Boltzmann machines (Salakhutdinov and Hinton, 2009a)
5. Unsupervised convolutional network (Jarrett *et al.*, 2009b)
6. GPU-accelerated multilayer perceptron (Ciresan *et al.*, 2010)
7. Distributed autoencoder (Le *et al.*, 2012)
8. Multi-GPU convolutional network (Krizhevsky *et al.*, 2012a)
9. COTS HPC unsupervised convolutional network (Coates *et al.*, 2013)
10. GoogLeNet (Szegedy *et al.*, 2014a)

Fig: I. Goodfellow

More neurons now

1. Perceptron (Rosenblatt, 1958, 1962)
2. Adaptive Linear Element (Widrow and Hoff, 1960)
3. Neocognitron (Fukushima, 1980)
4. Early backpropagation network (Rumelhart *et al.*, 1986b)
5. Recurrent neural network for speech recognition (Robinson and Fallside, 1991)
6. Multilayer perceptron for speech recognition (Bengio *et al.*, 1991)
7. Mean field sigmoid belief network (Saul *et al.*, 1996)
8. LeNet-5 (LeCun *et al.*, 1998b)
9. Echo state network (Jaeger and Hass, 2004)
10. Deep belief network (Hinton *et al.*, 2006)
11. GPU-accelerated convolutional network (Chellapilla *et al.*, 2006)
12. Deep Boltzmann machines (Salakhutdinov and Hinton, 2009a)
13. GPU-accelerated deep belief network (Rains *et al.*, 2009)
14. Unsupervised convolutional network (Jarrett *et al.*, 2009b)
15. GPU-accelerated multilayer perceptron (Ciresan *et al.*, 2010)
16. OMP-1 network (Coates and Ng, 2011)
17. Distributed autoencoder (Le *et al.*, 2012)
18. Multi-GPU convolutional network (Krizhevsky *et al.*, 2012a)
19. COTS HPC unsupervised convolutional network (Coates *et al.*, 2013)
20. GoogLeNet (Szegedy *et al.*, 2014a)

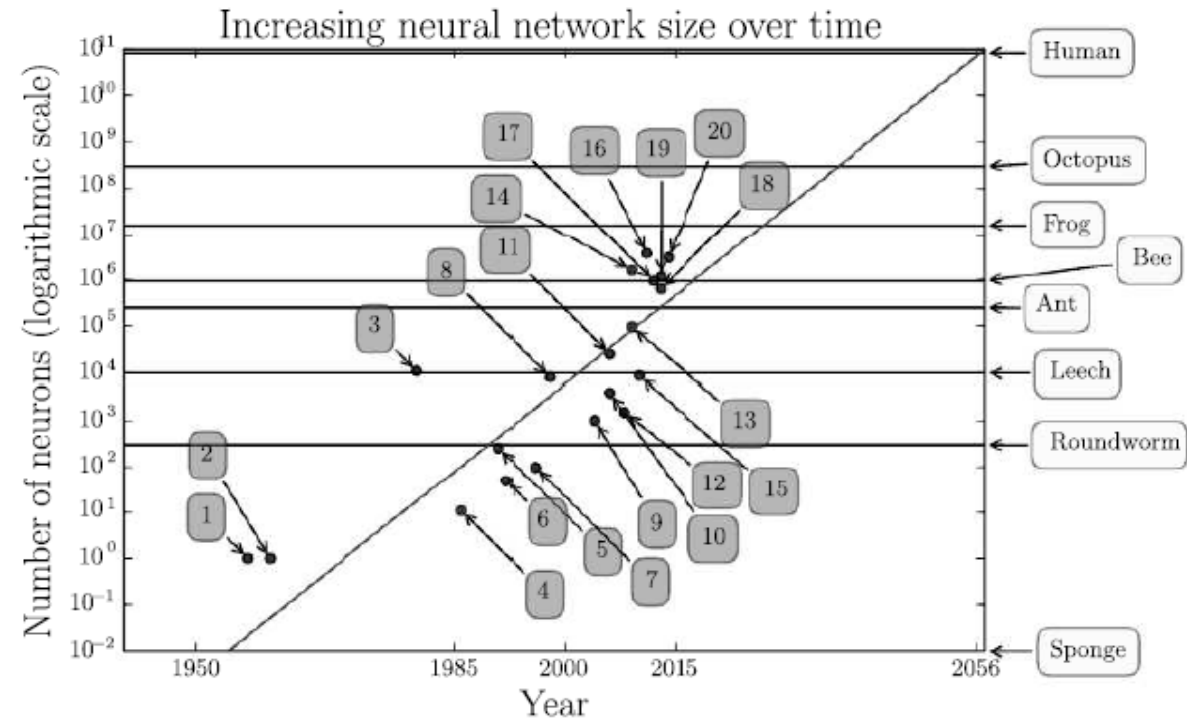


Fig: I. Goodfellow